

Neurotrophic factor Neurtin modulates T cell electrical and metabolic state for the balance of tolerance and immunity

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Abstract

The adaptive T cell response is accompanied by continuous rewiring of the T cell's electric and metabolic state. Ion channels and nutrient transporters integrate bioelectric and biochemical signals from the environment, setting cellular electric and metabolic states. Divergent electric and metabolic states contribute to T cell immunity or tolerance. Here, we report that neurtin (Nrn1) contributes to tolerance development by modulating regulatory and effector T cell function. Nrn1 expression in regulatory T cells promotes its expansion and suppression function, while expression in the T effector cell dampens its inflammatory response. Nrn1 deficiency causes dysregulation of ion channel and nutrient transporter expression in Treg and effector T cells, resulting in divergent metabolic outcomes and impacting autoimmune disease progression and recovery. These findings identify a novel immune function of the neurotrophic factor Nrn1 in regulating the T cell metabolic state in a cell context-dependent manner and modulating the outcome of an immune response.

eLife assessment

The neurotrophic factor Neuritin can moderate T-cell tolerance and immunity through both regulatory T (Treg) and effector T cells, promoting Treg cell expansion and suppression while dampening effector T cells to mediate the inflammatory response. Neuritin expression influences the membrane potential, ion channels, and nutrient transporter expression patterns of CD4⁺ T cells, contributing to differential metabolic states in Treg and effector T cells. These findings are **solid** and **important** for understanding immune regulation involving Treg cells and effector T cells.

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Introduction

Peripheral T cell tolerance is important in restricting autoimmunity and minimizing collateral damage during active immune reactions and is achieved via diverse mechanisms, including T cell anergy, regulatory T (Treg) cell mediated suppression, and effector T (Te) cell exhaustion or deletion (ElTanbouly and Noelle, 2021 [DOI](#)). Upon activation, Treg and conventional T cells integrate environmental cues and adapt their metabolism to the energetic and biosynthetic demands, leading to tolerance or immunity. Tolerized versus responsive T cells are characterized by differential metabolic states. For example, T cell anergy is associated with reduced glycolysis, whereas activated T effector cells exhibit increased glycolysis (Buck et al., 2017 [DOI](#); Geltink et al., 2018 [DOI](#); Peng and Li, 2023 [DOI](#); Zheng et al., 2009 [DOI](#)). Cellular metabolic states depend on electrolyte and nutrient uptake from the microenvironment (Chapman and Chi, 2022 [DOI](#); Olenchok et al., 2017 [DOI](#)). Ion channels and nutrient transporters, which can integrate environmental nutrient changes, affect the cellular metabolic choices and impact the T cell functional outcome (Babst, 2020 [DOI](#); Bohmwald et al., 2021 [DOI](#); Ramirez et al., 2018 [DOI](#)). Each cell's functional state would correspond with a set of ion channels and nutrient transporters supporting their underlying metabolic requirements. The mechanisms coordinating the ion channel and nutrient transporter expression changes to support the adaptive T cell functional state in the immune response microenvironment remain unclear.

Nrn1, also known as candidate plasticity gene 15 (CPG15), was initially discovered as a neurotrophic factor linked to the neuronal cell membrane through a glycosylphosphatidylinositol (GPI) anchor (Nedivi et al., 1998 [DOI](#); Zhou and Zhou, 2014 [DOI](#)). It is highly conserved across species, with 98% overall homology between the murine and human protein. Nrn1 plays multiple roles in neural development, synaptic plasticity, synaptic maturation, neuronal migration, and survival (Cantalops et al., 2000 [DOI](#); Javaherian and Cline, 2005 [DOI](#); Nedivi et al., 1998 [DOI](#); Putz et al., 2005 [DOI](#); Zito et al., 2014 [DOI](#)). In the immune system, Nrn1 expression has been found in Foxp3⁺ Treg and follicular regulatory T cells (Tfr) (Gonzalez-Figueroa et al., 2021 [DOI](#); Vahl et al., 2014 [DOI](#)), T cells from transplant tolerant recipients (Lim et al., 2013 [DOI](#)), anergized CD8 cells or CD8 cells from tumor-infiltrating lymphocytes in mouse tumor models (Schietinger et al., 2012 [DOI](#); Schietinger et al., 2016 [DOI](#); Singer et al., 2016 [DOI](#)), and in human Treg infiltrating breast cancer tumor tissue (Plitas et al., 2016 [DOI](#)). Soluble Nrn1 can be released from Tfr cells and act directly on B cells to suppress autoantibody development against tissue-specific antigens (Gonzalez-Figueroa et al., 2021 [DOI](#)). Despite the observation of Nrn1 expression in Treg cells and T cells from tolerant environments (Gonzalez-Figueroa et al., 2021 [DOI](#); Lim et al., 2013 [DOI](#); Plitas et al., 2016 [DOI](#); Schietinger et al., 2012 [DOI](#); Schietinger et al., 2016 [DOI](#); Singer et al., 2016 [DOI](#)), the roles of Nrn1 in T cell tolerance development and Treg cell function have not been explored, and functional mechanism of Nrn1 remains

elusive. This study demonstrates that the neurotrophic factor *Nrn1* can moderate T cell tolerance and immunity through both Treg and Te cells, impacting Treg cell expansion and suppression while controlling inflammatory response in Te cells.

Results

Nrn1 expression and function in T cell anergy

To explore the molecular mechanisms underlying peripheral tolerance development, we utilized a system we previously developed to identify tolerance-associated genes (Huang et al., 2004 [\[1\]](#)). We compared the gene expression patterns associated with either a T effector/memory response or tolerance induction triggered by the same antigen but under divergent *in vivo* conditions (Huang et al., 2004 [\[1\]](#)). Influenza hemagglutinin (HA) antigen-specific TCR transgenic CD4 T cells were adoptively transferred into WT recipients with subsequent HA-Vaccinia virus (VacHA) infection to generate T effector/memory cells while tolerogenic HA-specific CD4s were generated by transfer into hosts with transgenic expression of HA as self-antigen (C3-HA mice, **Figure 1A** [\[1\]](#)). (Huang et al., 2004 [\[1\]](#)). One of the most differentially expressed genes upregulated in the anergy-inducing condition was *Nrn1*. *Nrn1* expression was significantly higher among cells recovered from C3-HA hosts vs. cells from VacHA infected mice at all time points tested by qRT-PCR (**Figure 1A** [\[1\]](#)). To further confirm the association of *Nrn1* expression with T cell anergy, we assessed *Nrn1* expression in naturally occurring anergic polyclonal CD4⁺ T cells (Ta), which can be identified by surface co-expression of Folate Receptor 4 (FR4) and the ecto-5'-nucleotidase CD73 (Ta, CD4⁺CD44⁺FR4^{hi}CD73^{hi} cells) (Kalekar et al., 2016 [\[2\]](#)). *Nrn1* expression was significantly higher in Ta than in naïve CD4 (Tn, CD4⁺CD62L⁺CD44⁺FR4⁺CD73⁺) and antigen-experienced cells (Te, CD4⁺CD44⁺FR4⁺CD73⁺) under steady-state conditions measured by both qRT-PCR and western blot (**Figure 1B** [\[1\]](#)). Given that Treg cells, like anergic cells, have roles in maintaining immune tolerance, we queried whether *Nrn1* is also expressed in Treg cells. *Nrn1* expression can be detected in nTreg and induced Treg (iTreg) cells generated *in vitro* (**Figure 1C** [\[1\]](#)). To evaluate *Nrn1* expression under pathological tolerant conditions (Cuenca et al., 2003 [\[3\]](#)), we evaluated *Nrn1* expression in T cells within the tumor microenvironment. *Nrn1* expression in murine Treg cells and non-Treg CD4⁺ cells from tumor infiltrates were compared to the Treg cells and non-Treg CD4⁺ T cells isolated from peripheral blood. *Nrn1* mRNA level was significantly increased among tumor-associated Treg cells and non-Treg CD4 cells compared to cells from peripheral blood (Figure 1-figure supplement 1A). Consistent with our findings in the mouse tumor setting, the Treg and non-Treg T cells from human breast cancer infiltrates reveal significantly higher *Nrn1* expression compared to the peripheral blood Treg and non-Treg cells (Figure 1-figure supplement 1B) (Plitas et al., 2016 [\[4\]](#)).

CD4⁺ T cells may pass through an effector stage after activation before reaching an anergic state (Adler et al., 1998 [\[5\]](#); Chen et al., 2004 [\[6\]](#); Huang et al., 2003 [\[7\]](#); Opejin et al., 2020 [\[8\]](#)). To evaluate the potential role of *Nrn1* expression in T cell tolerance development, we further examined *Nrn1* expression kinetics after T cell activation. *Nrn1* expression was significantly induced after CD4⁺ T cell activation (**Figure 1D** [\[1\]](#)). Using a *Nrn1* specific antibody, *Nrn1* can be detected on activated CD4⁺ and CD8⁺ cells (**Figure 1D** [\[1\]](#), Figure 1-figure supplement 1C). The significant enhancement of *Nrn1* expression after T cell activation suggests that *Nrn1* may contribute to the process of T cell tolerance development and/or maintenance.

To understand the functional implication of *Nrn1* expression in immune tolerance, we analyzed *Nrn1*-deficient (*Nrn1*^{-/-}) mice (Fujino et al., 2011 [\[9\]](#)). In the first evaluation of the *Nrn1*^{-/-} colony, *Nrn1*^{-/-} mice had consistently reduced body weight compared to heterozygous *Nrn1*^{+/-} and WT (*Nrn1*^{+/+}) mice (Figure 1-figure supplement 2A). The lymphoid tissues of *Nrn1*^{-/-} mice were comparable to their *Nrn1*^{+/-} and WT counterparts except for a slight reduction in cell number that was observed in the spleens of *Nrn1*^{-/-} mice, likely due to their smaller size (Figure 1-figure

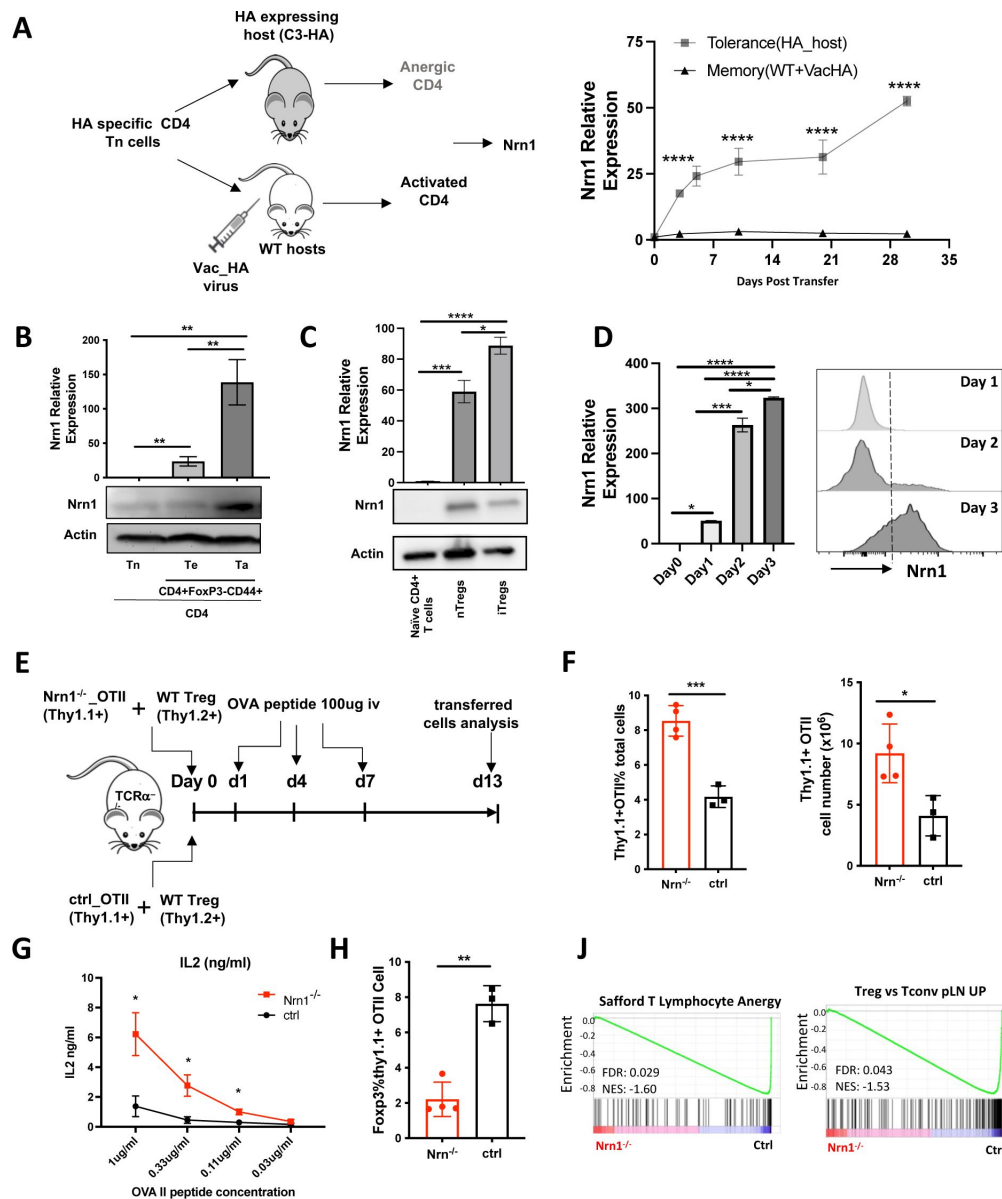


Figure 1.

Nrn1 expression and function in anergic T cells.

(A) Experimental scheme identifying Nrn1 in anergic T cells and qRT-PCR confirmation of Nrn1 expression in HA-specific CD4 cells recovered from HA-expressing host vs WT host activated with Vac_HA virus. (B) qRT-PCR and western blot detecting Nrn1 expression in naïve CD4⁺CD62L^{hi}CD44^{lo} Tn cell, CD4⁺Foxp3⁺CD44^{hi}CD73⁺FR⁺ Te cells and CD4⁺Foxp3⁺CD44^{hi}CD73⁺FR⁺ Ta cells. (C) Nrn1 expression was measured by qRT-PCR and western blot among naïve CD4⁺ T cells, CD4⁺Foxp3⁺ nTreg, and *in vitro* generated iTregs. (D) Nrn1 expression was detected by qRT-PCR and flow cytometry among naïve CD4⁺ cells and activated CD4⁺ cells on days 1, 2, and 3 after activation. qPCR Data are presented as average $\pm$ SEM. *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. Triplicates were used. Ordinary one-way ANOVA was performed for multi-comparison. (E-J). Anergy induction *in vivo*. (E) Experimental outline evaluating anergy development *in vivo*: 2x10⁶ Thy1.1⁺ Nrn1^{-/-} or ctrl CD4 OTII T cells were co-transferred with 5x10⁵ Thy1.2⁺Thy1.1⁻ WT Treg cells into TCR α ^{-/-} mice. Cells were recovered on day 13 post-transfer. (F) Proportions and numbers of OTII cells recovered from recipient spleen; (G) IL2 secretion from OTII cells upon *ex vivo* stimulation with OVA peptide. (H) Foxp3⁺ cell proportion among Thy1.1⁺ Nrn1^{-/-} or ctrl CD4 cells. (I & J) Nrn1^{-/-} vs ctrl OTII cells recovered from the peptide-induced anergy model were subjected to bulk RNASeq analysis. GSEA comparing the expression of signature genes for anergy (I) and Treg (J) among ctrl and Nrn1^{-/-} OTII cells. Data are presented as mean $\pm$ SEM and representative of 3 independent experiments (N $\geq$ 4 mice per group). *p<0.05, **p<0.01, ***p<0.001. Unpaired Student's t-tests were performed.

supplement 2B). Analysis of thymocytes revealed no defect in T cell development (Figure 1-figure supplement 2C), and a flow cytometric survey of the major immune cell populations in the peripheral lymphoid tissue of these mice revealed similar proportions of CD4, CD8 T cells, B cells, monocytes and dendritic cells (DCs) (Figure 1-figure supplement 2D). Similarly, no differences were found between the proportions of anergic and Treg cells in $Nrn1^{-/-}$, $Nrn1^{+/-}$ and WT mice (Figure 1-figure supplement 2E, F), suggesting that $Nrn1$ deficiency does not significantly affect anergic and Treg cell balance under steady state. Additionally, histopathology assessment of lung, heart, liver, kidney, intestine and spleen harvested from 13 months old $Nrn1^{-/-}$ and $Nrn1^{+/-}$ littermates did not reveal any evidence of autoimmunity (data not shown). The comparable level of anergic and Treg cell population among $Nrn1^{-/-}$, $Nrn1^{+/-}$ and WT mice and lack of autoimmunity in $Nrn1^{-/-}$ aged mice suggest that $Nrn1$ deficiency is not associated with baseline immune abnormalities or overt dysfunction. Due to the similarity between $Nrn1^{+/-}$ and WT mice, we have used either $Nrn1^{+/-}$ or WT mice as our control depending on mice availability and referred to both as “ctrl” in the subsequent discussion.

To evaluate the relevance of $Nrn1$ in CD4⁺ T cell tolerance development, we employed the classic peptide-induced T cell anergy model (Vanasek et al., 2006 [\[1\]](#)). Specifically, we crossed OVA antigen-specific TCR transgenic OTII mice onto the $Nrn1^{-/-}$ background. $Nrn1^{-/-}$ _OTII⁺ or control_OTII⁺ (ctrl_OTII⁺) cells marked with Thy1.1⁺ congenic marker (Thy1.1⁺Thy1.2⁻), were co-transferred with polyclonal WT Tregs (marked as Thy1.1⁺Thy1.2⁺), into TCRα knockout mice (TCRα^{-/-}), followed by injection of soluble OVA peptide to induce clonal anergy (**Figure 1E** [\[1\]](#)) (Chappert and Schwartz, 2010 [\[2\]](#); Martinez et al., 2012 [\[3\]](#); Mercadante and Lorenz, 2016 [\[4\]](#); Shin et al., 2014 [\[5\]](#)). On day 13 after cell transfer, the proportion and number of OTII cells increased in the $Nrn1^{-/-}$ _OTII compared to the ctrl_OTII hosts (**Figure 1F** [\[1\]](#)). Moreover, $Nrn1^{-/-}$ _OTII cells produced increased IL2 than ctrl_OTII upon restimulation (**Figure 1G** [\[1\]](#)). Anergic CD4 Tconv cells can transdifferentiate into Foxp3⁺ pTreg cells *in vivo* (DL, 2017; Kalekar et al., 2016 [\[6\]](#); Kuczma et al., 2021 [\[7\]](#)). Consistent with reduced anergy induction, the proportion of Foxp3⁺ pTreg among $Nrn1^{-/-}$ _OTII was significantly reduced (**Figure 1H** [\[1\]](#)). In parallel with the phenotypic analysis, we compared gene expression between $Nrn1^{-/-}$ _OTII and ctrl_OTII cells by RNA Sequencing (RNASeq). Gene set enrichment analysis (GSEA) revealed that the gene set on T cell anergy was enriched in ctrl relative to $Nrn1^{-/-}$ _OTII cells (**Figure 1I** [\[1\]](#)) (Safford et al., 2005 [\[8\]](#)). Also, consistent with the decreased transdifferentiation to Foxp3⁺ cells, the Treg signature gene set was prominently reduced in $Nrn1^{-/-}$ _OTII cells relative to the ctrl (**Figure 1J** [\[1\]](#)). Anergic T cells are characterized by inhibition of proliferation and compromised effector cytokines such as IL2 production (Choi and Schwartz, 2007 [\[9\]](#)). The increased cell expansion and cytokine production in $Nrn1^{-/-}$ _OTII cells and the reduced expression of anergic and Treg signature genes all support the notion that $Nrn1$ is involved in T cell anergy development.

Anergic T cells are developed after encountering antigen, passing through a brief effector stage, and reaching an anergic state (Chappert and Schwartz, 2010 [\[2\]](#); Huang et al., 2003 [\[10\]](#); Silva Morales and Mueller, 2018 [\[11\]](#); Zha et al., 2006 [\[12\]](#)). Enhanced T cell activation, defective Treg cell conversion or expansion, and heightened T effector cell response may all contribute to defects in T cell anergy induction and/or maintenance (Chappert and Schwartz, 2010 [\[2\]](#); Huang et al., 2003 [\[10\]](#); Kalekar et al., 2016 [\[6\]](#); Silva Morales and Mueller, 2018 [\[11\]](#); Zha et al., 2006 [\[12\]](#)). We first examined early T cell activation to understand the underlying cause of defective anergy development in $Nrn1^{-/-}$ cells. $Nrn1^{-/-}$ CD4⁺ cells showed reduced T cell activation, as evidenced by reduced CellTrace violet dye (CTV) dilution, activation marker expression, and Ca⁺⁺ entry after TCR stimulation (Figure 1-figure supplement 3A, B, C). The reduced early T cell activation observed in $Nrn1^{-/-}$ CD4 cells suggests that the compromised anergy development in $Nrn1^{-/-}$ _OTII cells was not caused by enhanced early T cell activation. The defective pTreg generation and/or enhanced effector T cell response may contribute to compromised anergy development.

Compromised Treg expansion and suppression in the absence of *Nrn1*

The significant reduction of Foxp3⁺ pTreg among *Nrn1*^{-/-} OTII cells could be caused by the diminished conversion of Foxp3⁻ Tconv cells to pTreg and/or diminished Treg cell expansion and persistence. To understand the cause of pTreg reduction in *Nrn1*^{-/-} OTII cells (**Figure 1H**), we turned to the induced Treg (iTreg) differentiation system to evaluate the capability of Foxp3⁺ Treg development and expansion in *Nrn1*^{-/-} cells. Similar proportions of Foxp3⁺ cells were observed in *Nrn1*^{-/-} and ctrl cells under the iTreg culture condition, suggesting that *Nrn1* deficiency does not significantly impact Foxp3⁺ cell differentiation. To examine the capacity of iTreg expansion, *Nrn1*^{-/-} and ctrl iTreg cells were restimulated with anti-CD3, and we found reduced live cells over time in *Nrn1*^{-/-} iTreg compared to the ctrl (**Figure 2A**). The reduced live cell number in *Nrn1*^{-/-} was accompanied by reduced Ki67 expression (**Figure 2B**). Although *Nrn1*^{-/-} iTregs retained a higher proportion of Foxp3⁺ cells three days after restimulation, however, when taking into account the total number of live cells, the actual number of live Foxp3⁺ cells was reduced in *Nrn1*^{-/-} (**Figure 2C**). Treg cells are not stable and are prone to losing Foxp3 expression after extended proliferation (Feng et al., 2014; Floess et al., 2007; Li et al., 2014; Zheng et al., 2010). The increased proportion of Foxp3⁺ cells was consistent with reduced proliferation observed in *Nrn1*^{-/-} cells. Thus, *Nrn1* deficiency can lead to reduced iTreg cell proliferation and persistence *in vitro*.

The defects observed in iTreg cell expansion *in vitro* prompt further examination of *Nrn1*^{-/-} nTreg expansion and suppression function *in vivo*. To this end, we tested the suppression capacity of congenically marked (CD45.1⁺CD45.2⁺) *Nrn1*^{-/-} or ctrl nTreg toward CD45.1⁺CD45.2⁻ responder cells in Rag2^{-/-} mice (**Figure 2D**). The CD45.1⁺CD45.2⁻ responder cells devoid of Treg cells were splenocytes derived from Foxp3DTRGFP (FDG) mice pretreated with diphtheria toxin (DT) (Kim et al., 2007; Workman et al., 2011). DT treatment caused the deletion of Treg cells in FDG mice (Kim et al., 2007). Although the CD45.1⁺CD45.2⁺ *Nrn1*^{-/-} and ctrl cell proportions were not significantly different among hosts splenocytes at day 7 post transfer (**Figure 2E**), *Nrn1*^{-/-} cells retained a higher Foxp3⁺ cell proportion and reduced Ki67 expression comparing to the ctrl (**Figure 2F**, G). These findings were similar to our observation of iTreg cells *in vitro* (**Figure 2B**, C). *Nrn1*^{-/-} Tregs also showed reduced suppression toward CD45.1⁺ responder cells, evidenced by increased CD45.1⁺ proportion and cell number in host splenocytes (**Figure 2H**).

To evaluate the functional implication of *Nrn1*^{-/-} Treg suppression in disease settings, we challenged the Rag2^{-/-} hosts with the poorly immunogenic B16F10 tumor (**Figure 2D**). Tumors grew much slower in *Nrn1*^{-/-} Treg recipients than those reconstituted with ctrl Tregs (**Figure 2I**). Moreover, the number of CD45.1⁺ cells in tumor-draining lymph nodes and spleens increased significantly in *Nrn1*^{-/-} Treg hosts compared to the ctrl group (**Figure 2J**). Consistently, the CD45.1⁺ responder cell proportion among tumor lymphocyte infiltrates (TILs) was also increased (**Figure 2K**), accompanied by an increased proportion of IFNγ⁺ cells among CD8 TILs from *Nrn1*^{-/-} Treg hosts (**Figure 2L**). The increased expansion of CD45.1⁺ responder cells and reduced tumor growth further confirmed the reduced suppressive capacity of *Nrn1*^{-/-} Treg cells.

Nrn1 impacts Treg cell electrical and metabolic state

To understand the molecular mechanisms associated with *Nrn1*^{-/-} Treg cells, we compared gene expression between *Nrn1*^{-/-} and ctrl iTregs under resting (IL2 only) and activation (aCD3 and IL2) conditions by RNASeq. GSEA on gene ontology database and clustering of enriched gene sets by Cytoscape identified three clusters enriched in resting *Nrn1*^{-/-} iTreg (**Figure 3A**, Figure 3-figure supplement Table 1) (Shannon et al., 2003; Subramanian et al., 2005). The “neurotransmitter involved in membrane potential (MP)” and “sodium transport” clusters involved gene sets on the ion transport and cell MP regulation (**Figure 3A**, Figure 3-figure supplement Table 1). MP is the difference in electric charge between the interior and the exterior of the cell membrane (Abdul

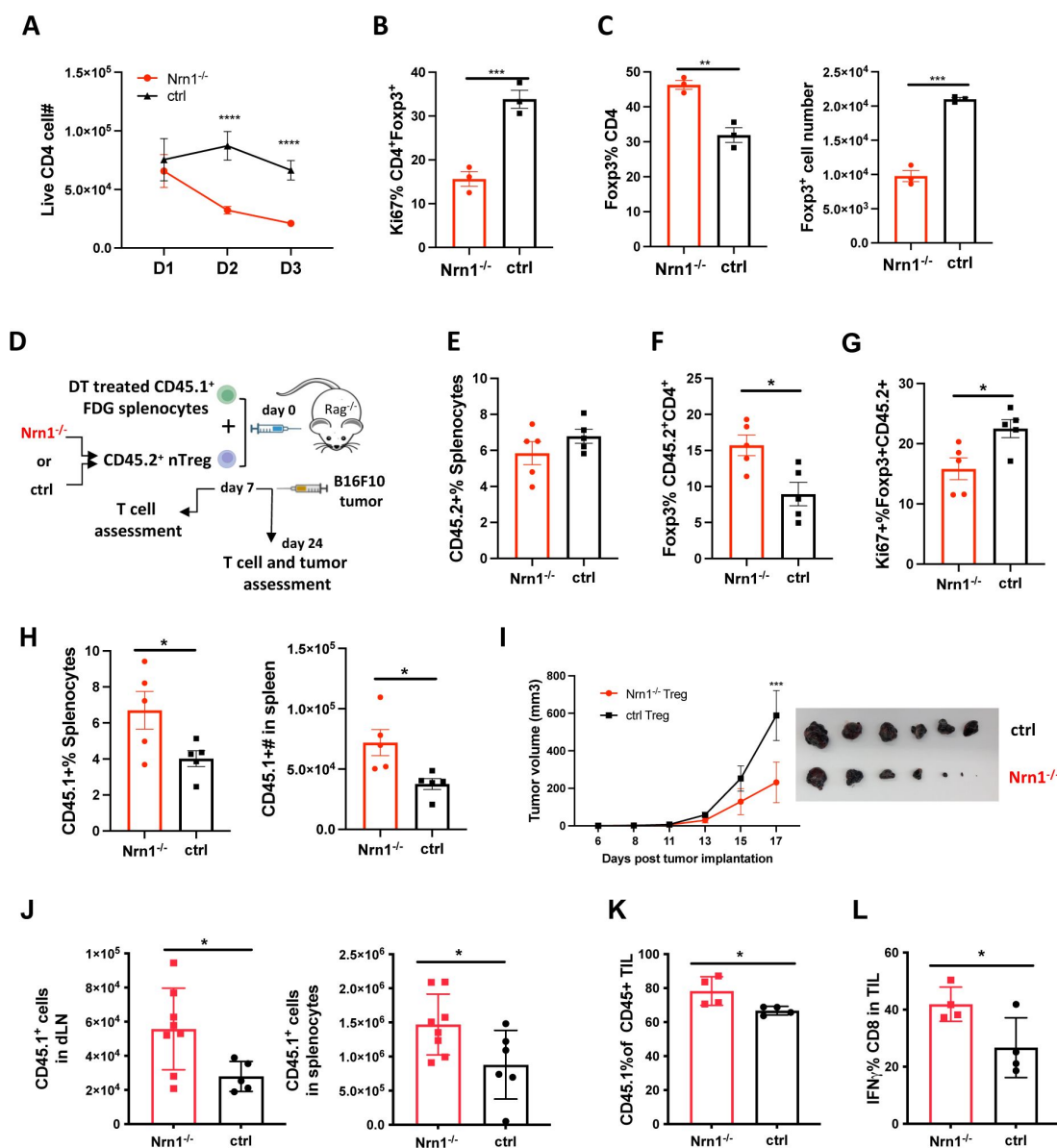


Figure 2.

Reduced proliferation and suppression function in *Nrn1*^{-/-} Treg cells.

(A-C) iTreg cell expansion after restimulation. (A) The number of live cells from day 1 to day 3 after iTreg cell restimulation with anti-CD3. (B) Ki67 expression among CD4⁺Foxp3⁺ cells day 3 after restimulation. (C) Foxp3⁺ cell proportion and number among live CD4⁺ cells day 3 after restimulation. Triplicates in each experiment, data represent one of four independent experiments. (D-L). *Nrn1*^{-/-} or ctrl nTreg cells expansion and suppression *in vivo*. (D) The experimental scheme. CD45.2⁺ nTreg T cells from *Nrn1*^{-/-} or ctrl were transferred with CD45.1⁺ FDG splenocytes devoid of Tregs into Rag2^{-/-} host. Treg cell expansion and suppression toward FDG CD45.1⁺ responder cells were evaluated on day 7 post cell transfer. Alternatively, B16F10 tumor cells were inoculated on day 7 after cell transfer and monitored for tumor growth. (E-H) CD45.2⁺ cell proportion (E), Foxp3 retention (F), and Ki67 expression among Foxp3⁺ cells (G) at day 7 post cell transfer. (H) CD45.1⁺ cell proportion and number in the spleen of *Nrn1*^{-/-} or ctrl Treg hosts day 7 post cell transfer. (I-L). Treg cell suppression toward anti-tumor response. (I) Tumor growth curve and tumor size at harvest from *Nrn1*^{-/-} or ctrl nTreg hosts. (J) CD45.1⁺ cell count in tumor draining lymph node (LN) and spleen. (K) the proportion of CD45.1⁺ cells among CD45⁺ tumor lymphocyte infiltrates (TILs). (L) IFN γ % among CD8⁺ T cells in TILs. $n \geq 5$ mice per group. (E-H) represents three independent experiments, (I-L) represents two independent experiments. Data are presented as mean $\pm$ SEM * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$. Unpaired Student's t-tests were performed.

Kadir et al., 2018 [\[1\]](#); Blackiston et al., 2009 [\[2\]](#); Ma et al., 2017 [\[3\]](#)). Ion channels and transporters for Na^+ and other ions such as K^+ , Cl^- et al. maintain the ion balance and contribute to cell MP (Blackiston et al., 2009 [\[2\]](#)). MP change can impact cell plasma membrane lipid dynamics and affect receptor kinase activity (Zhou et al., 2015 [\[4\]](#)). The enrichment of “receptor protein kinase” gene set clusters may reflect changes caused by MP (**Figure 3A** [\[5\]](#), Figure 3-figure supplement Table 1). Gene set cluster analysis on activated iTreg cells also revealed the enrichment of the “ion channel and receptor” cluster in $\text{Nrnr1}^{-/-}$ cells (**Figure 3B** [\[6\]](#), Figure 3-figure supplement table 2), supporting the potential role of Nrnr1 in modulating ion balances and MP.

The “Neurotransmitter receptor activity involved in regulation of postsynaptic membrane potential” gene set was significantly enriched under resting and activation conditions (**Figure 3C and D** [\[7\]](#); Figure 3-figure supplement Table 3). The α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor (AMPA) subunits *Gria2* and *Gria3* are the major components of this gene set and showed increased expression in $\text{Nrnr1}^{-/-}$ cells (**Figure 3D** [\[8\]](#)). AMPA is an ionotropic glutamate receptor that mediates fast excitatory synaptic transmission in neurons. Nrnr1 has been reported as an accessory protein for AMPA (*Pandya et al., 2018* [\[9\]](#); *Schwenk et al., 2012* [\[10\]](#); *Subramanian et al., 2019* [\[11\]](#)), although the functional implication of Nrnr1 as an AMPA accessory protein remains unclear. The enrichment of MP related gene set prompted the examination of electric status, including MP level and ion channel expressions. We examined the relative MP level by FLIPR MP dye, a lipophilic dye able to cross the plasma membrane, which has been routinely used to measure cell MP changes (*Dvorak et al., 2021* [\[12\]](#); *Joesch et al., 2008* [\[13\]](#); *Nik et al., 2017* [\[14\]](#); *Whiteaker et al., 2001* [\[15\]](#)). When the cells are depolarized, the dye enters the cells, causing an increase in fluorescence signal. Conversely, cellular hyperpolarization results in dye exit and decreased fluorescence. Compared to ctrl iTreg cells, $\text{Nrnr1}^{-/-}$ exhibits significant hyperpolarization under both resting and activation conditions (**Figure 3E** [\[16\]](#)). Consistent with the MP change, the “MF_metal ion transmembrane transporter activity” gene set, which contains 436 ion channel related genes, was significantly enriched and showed a different expression pattern in $\text{Nrnr1}^{-/-}$ iTregs (**Figure 3F** [\[17\]](#); Figure 3-figure supplement 1A and B). The changes in cellular MP and differential expression of ion channel and transporter genes in $\text{Nrnr1}^{-/-}$ implicate the role of Nrnr1 in the balance of electric state in the iTreg cell.

MP changes have been associated with changes in amino acid (AA) transporter expression and nutrient acquisition, which in turn influences cellular metabolic and functional state (*Yu et al., 2022* [\[18\]](#)). To understand whether MP changes in $\text{Nrnr1}^{-/-}$ are associated with changes in nutrient acquisition and thus the metabolic state, we surveyed AA transport-related gene expression using the “Amino acid transmembrane transporter activity” gene set and found differential AA transporter gene expression between $\text{Nrnr1}^{-/-}$ and ctrl iTregs (**Figure 3G** [\[19\]](#)). Electrolytes and AAs entry are critical regulators of mTORC1 activation and T cell metabolism (*Liu and Sabatini, 2020* [\[20\]](#); *Saravia et al., 2020* [\[21\]](#); *Sinclair et al., 2013* [\[22\]](#); *Wang et al., 2020* [\[23\]](#)). We examined mTORC1 activation at the protein level by evaluating mTOR and S6 phosphorylation via flow cytometry. We found reduced phosphorylation of mTOR and S6 in activated $\text{Nrnr1}^{-/-}$ iTreg cells (**Figure 3H** [\[24\]](#)). We further performed a nutrient-sensing assay to evaluate the role of ion and nutrient entry in mTORC1 activation. $\text{Nrnr1}^{-/-}$ and ctrl iTreg cells were starved for one hour in a nutrient-free buffer, followed by adding RPMI medium with complete ions and nutrients, and cultured for two more hours. While adding the medium with nutrients clearly increased the mTOR and S6 phosphorylation, the degree of change was significantly less in $\text{Nrnr1}^{-/-}$ than in the ctrl (**Figure 3I** [\[25\]](#)). Consistently, GSEA on Hallmark gene sets reveal reduced gene set enrichment relating to the mTORC1 signaling, corroborating the reduced pmTOR and pS6 detection in $\text{Nrnr1}^{-/-}$ cells. Moreover, $\text{Nrnr1}^{-/-}$ cells also showed reduced expression of glycolysis, fatty acid metabolism, and oxidative phosphorylation related gene sets under both resting and activating conditions (**Figure 3J** [\[26\]](#), Figure 3-figure supplement 1C), indicating changes in metabolic status. Since previous work has identified mTORC1 to be an important regulator of aerobic glycolysis and given that our GSEA data suggested changes in glycolysis (**Figure 3J** [\[26\]](#)) (*Salmond, 2018* [\[27\]](#)), we performed the seahorse assay and confirmed reduced glycolysis among $\text{Nrnr1}^{-/-}$ cells (**Figure 3K** [\[28\]](#)). Examination of

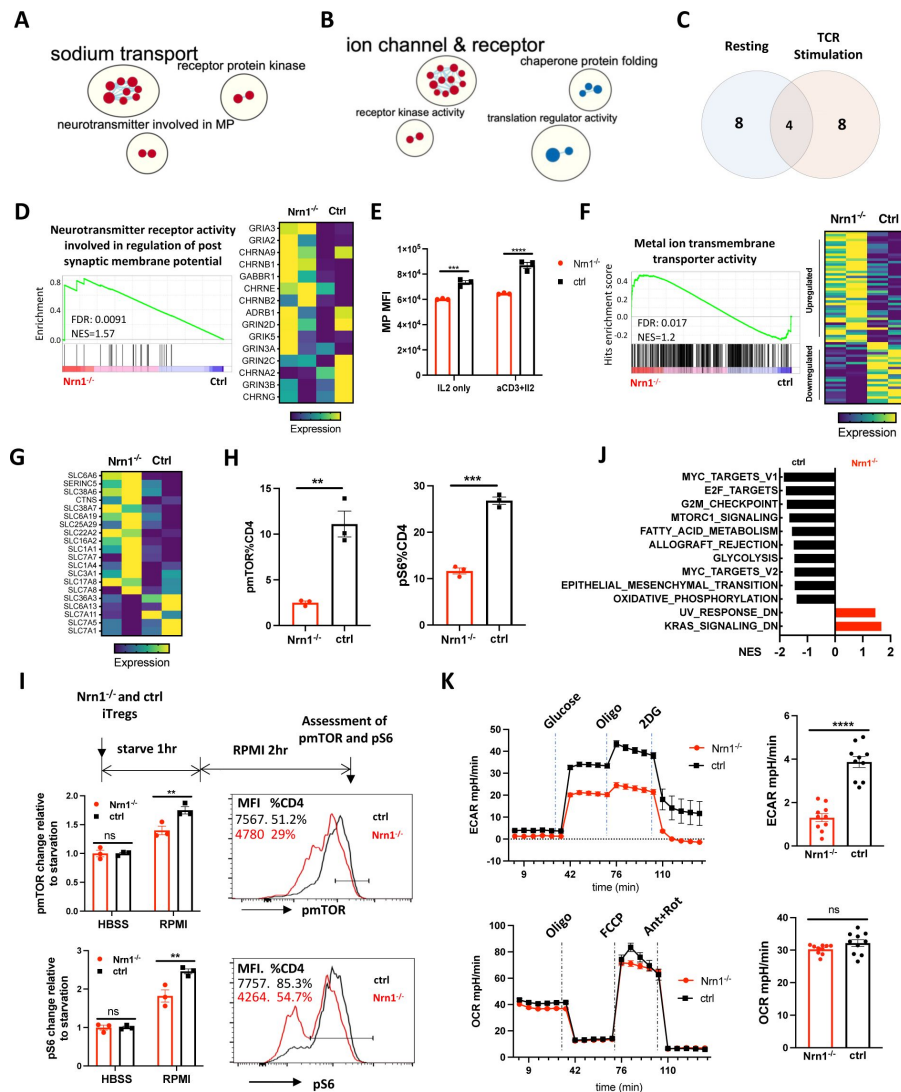


Figure 3.

Nrn1 expression impacts Treg cell electrical and metabolic state.

(A-C). Gene sets clusters enriched in *Nrn1*^{-/-} and ctrl iTreg cells. Gene sets cluster analysis via Cytoscape was performed on Gene ontology Molecular Function (GO_MF) gene sets. The results cutoff: p-value ≤ 0.05 and FDR q-value ≤ 0.1 . (A) Gene sets cluster in *Nrn1*^{-/-} iTreg cells cultured under resting condition (IL2 only) (Figure 3-figure supplement Table 1). (B) Gene sets clusters in *Nrn1*^{-/-} and ctrl iTreg cells reactivated with anti-CD3 (Figure 3-figure supplement Table 2). (C) Comparison of enriched gene sets in *Nrn1*^{-/-} under resting vs. activating condition (Figure 3-figure supplement Table 3). (D-F) Changes relating to cell electric state. (D) Enrichment of "GOMF_Neurotransmitter receptor activity involved in the regulation of postsynaptic membrane potential" gene set and enriched gene expression heatmap. (E) Membrane potential was measured in *Nrn1*^{-/-} and ctrl iTreg cells cultured in IL2 or activated with anti-CD3 in the presence of IL2. Data represent three independent experiments. (F) Enrichment of "GOMF_Metal ion transmembrane transporter activity" gene set and enriched gene expression heatmap (Figure 3- figure supplement 1A). (G-K). Metabolic changes associated with *Nrn1*^{-/-} iTreg. (G) Heatmap of differentially expressed amino acid (AA) transport-related genes (from "MF_Amino acid transmembrane transporter activity" gene list) in *Nrn1*^{-/-} and ctrl iTreg cells. (H) pmTOR and pS6 levels in TCR activated iTreg cells measured by flow cytometry. n=3 replicates per group. Data represent three independent experiments. (I) Measurement of pmTOR and pS6 in iTreg cells that were deprived of nutrients for 1h and refed with RPMI for two hours. (J) Hallmark gene sets significantly enriched in *Nrn1*^{-/-} and ctrl iTreg. NOM p-val<0.05, FDR q-val<0.25. (K) Seahorse analysis of extracellular acidification rate (ECAR) and oxygen consumption rate (OCR) in *Nrn1*^{-/-} and ctrl iTreg cells. n=6~10 technical replicates per group. Data represent three independent experiments. **p<0.01, ***p<0.001, ****p<0.0001. Unpaired student t-test for two-group comparison. Unpaired t-test (H, K), two-way ANOVA (E, I). ns, not significant.

mitochondrial bioenergetic function revealed a similar oxygen consumption rate (OCR) between $Nrn1^{-/-}$ and ctrl cells (**Figure 3K**). Thus, $Nrn1$ expression can affect the iTreg electric state, influence ion channel and nutrient transporter expression, impact nutrient sensing, modulate metabolic state, and contribute to Treg expansion and suppression function.

$Nrn1$ impact effector T cell inflammatory response

$CD4^{+}$ T cells can pass through an effector stage on their way to an anergic state (Huang et al., 2003). Since $Nrn1$ expression is significantly induced after T cell activation (**Figure 1D**), $Nrn1$ might influence $CD4^{+}$ effector (Te) cell differentiation, affecting anergy development. $Nrn1$ may exert different electric changes due to distinct ion channel expression contexts in Te cells than in Tregs. We first evaluated $Nrn1^{-/-}$ Te cell differentiation *in vitro*. $Nrn1$ deficient $CD4$ Te cells showed increased Ki67 expression, associated with increased cytokine TNF α , IL2, and IFN γ expression upon restimulation (**Figure 4A**). To evaluate $Nrn1^{-/-}$ Te cell response *in vivo*, we crossed $Nrn1^{-/-}$ with FDG mice and generated $Nrn1^{-/-}$ FDG and ctrl_FDG mice, which enabled the elimination of endogenous Treg cells (**Figure 4B**). Deleting endogenous Foxp3 $^{+}$ Treg cells using DT will cause the activation of self-reactive T cells, leading to an autoimmune response (Kim et al., 2007; Nystrom et al., 2014). Upon administration of DT, we observed accelerated weight loss in $Nrn1^{-/-}$ FDG mice, reflecting enhanced autoimmune inflammation (**Figure 4C**). Examination of T cell response revealed a significant increase in Ki67 expression and inflammatory cytokine TNF α , IL2, and IFN γ expression among $Nrn1^{-/-}$ $CD4$ cells on day 6 post DT treatment (**Figure 4D**), consistent with the findings *in vitro*. The proportion of Foxp3 $^{+}$ cells was very low on day 6 post DT treatment and comparable between $Nrn1^{-/-}$ and the ctrl (**Figure 4E**), suggesting that the differential Te cell response was not due to the impact from Treg cells. Thus, $Nrn1$ deficiency enhances Te cell response *in vitro* and *in vivo*.

To identify molecular changes responsible for $Nrn1^{-/-}$ Te phenotype, we compared gene expression between $Nrn1^{-/-}$ and ctrl Te cells by RNASeq. GSEA and Cytoscape analysis identified a cluster of gene sets on “membrane repolarization”, suggesting that $Nrn1$ may also be involved in the regulation of MP under Te context (**Figure 4F**, Figure 4-figure supplement Table 4) (Shannon et al., 2003; Subramanian et al., 2005). While the “membrane_repolarization” gene set was enriched in $Nrn1^{-/-}$ (**Figure 4G**), the “Neurotransmitter receptor activity involved in regulation of postsynaptic membrane potential” gene set was no longer enriched, but the AMPAR subunit Gria3 expression was still elevated in $Nrn1^{-/-}$ Te cells (Figure 4-figure supplement 1A). Although MP in Te cells was comparable between $Nrn1^{-/-}$ and ctrl (**Figure 4H**), the “MF_metal ion transmembrane transporter activity” gene set was significantly enriched in $Nrn1^{-/-}$ with different gene expression patterns (**Figure 4I**, Figure 4-figure supplement 1B), indicative of different electric state. The significant enrichment of ion channel related genes in $Nrn1^{-/-}$ Te cells was in line with the finding in $Nrn1^{-/-}$ iTreg cells, supporting the notion that $Nrn1$ expression may be involved in ion balance and MP modulation.

Examination of nutrient transporters revealed that the “Amino acid transmembrane transporter activity” gene set was significantly enriched in $Nrn1^{-/-}$ cells than the ctrl (**Figure 4J**). Along with the enrichment of ion channel and nutrient transporter genes (**Figure 4I and J**), we found enhanced mTOR and S6 phosphorylation in $Nrn1^{-/-}$ Te cells (**Figure 4K**). We also compared nutrient sensing capability between $Nrn1^{-/-}$ and ctrl Te cells, as outlined in **Figure 3I**. $Nrn1^{-/-}$ Te showed increased mTOR and S6 phosphorylation after sensing ions and nutrients in RPMI medium (**Figure 4L**), confirming the differential impact of ions and nutrients on $Nrn1^{-/-}$ and ctrl Te cells. GSEA on Hallmark collection showed enrichment of mTORC1 signaling gene set (**Figure 4M**), corroborating with increased pmTOR and pS6 detection in $Nrn1^{-/-}$ Te cells. Along with increased mTORC1 signaling, $Nrn1^{-/-}$ Te cells also showed enrichment of gene sets on glycolysis and proliferation (**Figure 4M**). Evaluation of metabolic changes by seahorse confirmed increased glycolysis in $Nrn1^{-/-}$ cells, while the OCR remained comparable between $Nrn1^{-/-}$ and ctrl

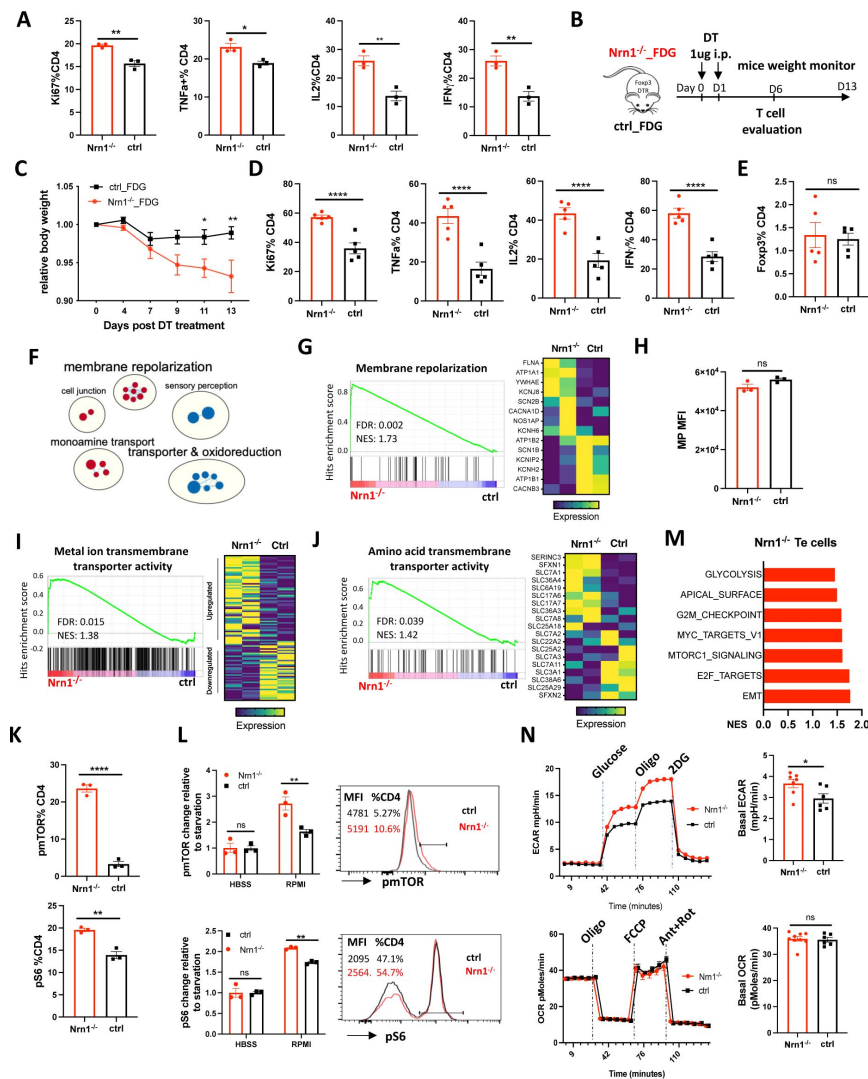


Figure 4.

Nrn1 deficiency affects Te cell response.

(A) Comparison of cell proliferation and cytokine expression in Nrn1^{-/-} and ctrl Te cells. Data represent one of three independent experiments. (B-E) An enhanced autoimmune response in Nrn1^{-/-} mice *in vivo*. (B) Experimental scheme. Nrn1^{-/-} mice were crossed with FDG mice and Nrn1^{-/-}_FDG or ctrl_FDG mice were obtained. The autoimmune response was induced by injecting DT i.p. to delete endogenous Treg cells. Mice's weight change was monitored after disease induction. (C) Relative body weight change after autoimmune response induction. (D) Mice were harvested 6 days after DT injection and assessed for ki67, cytokine TNFα, IL2, and IFNγ expression in CD4⁺ cells. (E) Foxp3 expression among CD4⁺ cells day 6 post DT treatment. n≥5 mice per group. Data represent four independent experiments. (F-I) Changes relating to ion balances in Te cells. (F) Gene sets clusters from GSEA of GO_MF and GO_Biological process (GO_BP) results in Nrn1^{-/-} and ctrl Te cells (Figure 4-figure supplement Table 4). (G) Enrichment of "GOBP_membrane repolarization" gene set and enriched gene expression heatmap. (H) Membrane potential measurement in Te cells. Data represent two independent experiments. (I) Enrichment of "GOMF_Metal ion transmembrane transporter activity" gene set and heatmap of differential gene expression pattern (Figure 4-figure supplement 1B). (J-N) Metabolic changes associated with Nrn1^{-/-} Te cell. (J) Enrichment of "GOMF_amino acid transmembrane transporter activity" gene set and differential gene expression heatmap. (K) Phosphorylation of mTOR and S6 in Te cells measured by flow cytometry. n=3 replicates per group. Data represent two independent experiments. (L) Measurement of pmTOR and pS6 in Te cells after nutrient sensing. Data represent three independent experiments. (M) Enriched Hallmark gene sets (p<0.05, FDR q<0.25). (N) Seahorse analysis of extracellular acidification rate (ECAR) and oxygen consumption rate (OCR) in Nrn1^{-/-} and ctrl Te cells. n≥6 technical replicates per group. Data represent three independent experiments. Error bars indicate ±SEM. *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001, unpaired Student's t-test was performed for two-group comparison.

(**Figure 4N** [↗](#)). These *in vitro* studies on Te cells indicate that *Nrn1* deficiency resulted in the dysregulation of the electrolyte and nutrient transport program, impacting Te cell nutrient sensing, metabolic state, and the outcome of inflammatory response.

Nrn1 deficiency exacerbates autoimmune disease

The coordinated reaction of Treg and Te cells contributes to the outcome of the immune response. We employed the experimental autoimmune encephalomyelitis (EAE), the murine model of multiple sclerosis (MS) to evaluate the overall impact of *Nrn1* on autoimmune disease development. Upon EAE induction, the incidence and time to EAE onset in *Nrn1*^{-/-} mice were comparable to the ctrl mice, but the severity, disease persistence, and body weight loss were increased in *Nrn1*^{-/-} mice (**Figure 5A** [↗](#)). Exacerbated EAE was associated with significantly increased CD45⁺ cell infiltrates, increased CD4⁺ cell number, increased proportion of MOG-specific CD4 cells, and reduced proportion of Foxp3⁺ CD4 cells in the *Nrn1*^{-/-} spinal cord (**Figure 5B-E** [↗](#)). Moreover, we also observed increased proportions of IFN7⁺ and IL17⁺ CD4 cells in *Nrn1*^{-/-} mice (**Figure 5F** [↗](#)). Thus, the results from EAE corroborated with earlier data and confirmed the important role of *Nrn1* in establishing immune tolerance and modulating autoimmunity.

Discussion

T cell expansion and functional development depend on adaptive electric and metabolic changes, maintaining electrolyte balances, and appropriate nutrient uptake. The negative charge of the plasma membrane, ion channel expression pattern, and function are key characteristics associated with the cellular electric state in different systems, impacting cell proliferation and function (Blackiston et al., 2009 [↗](#); Emmons-Bell and Hariharan, 2021 [↗](#); Kiefer et al., 1980 [↗](#); Monroe and Cambier, 1983 [↗](#); Sundelacruz et al., 2009 [↗](#)). The electrolytes and nutrients, including amino acids, metabolites, and small peptides transported through ion channels and nutrient transporters, are also regulators and signaling agents impacting the choice of cellular metabolic pathways and functional outcomes (Hamill et al., 2020 [↗](#)). In this study, we report that the neurotropic factor *Nrn1* expression influences CD4 T cell MP, ion channels, and nutrient transporter expression patterns, contributing to differential metabolic states in Treg and Te cells. *Nrn1* deficiency compromises Treg cell expansion and suppression while enhancing Te cell inflammatory response, exacerbating autoimmune disease.

Bioelectric controls have been defined as a type of epigenetics that can control information residing outside of genomic sequence (Levin, 2021 [↗](#)). The sum of ion channels and pump activity generates the ionic gradient across the cell membrane, establishing the MP level and bioelectric state. Cells with the same MP can have different ion compositions, and the same ion channel may have a differential impact on MP when in combination with different ion channels (Abdul Kadir et al., 2018 [↗](#)). Consistent with this notion, *Nrn1* deficiency has differential impacts on the cellular electric state under the Treg and Te cells with different ion channel combinations. Altered MP was detected in *Nrn1* deficient Treg cells (**Figure 3E** [↗](#)), while comparable MP was observed between *Nrn1*^{-/-} and ctrl Te cells (**Figure 4H** [↗](#)). The MP level determined by ion channels and pump activity can influence the nutrient transport pattern, establishing a metabolic and functional state matching the MP level (Blackiston et al., 2009 [↗](#); Emmons-Bell and Hariharan, 2021 [↗](#); Kiefer et al., 1980 [↗](#); Monroe and Cambier, 1983 [↗](#); Sundelacruz et al., 2009 [↗](#); Yu et al., 2022 [↗](#)). Yu et al. reported that macrophage MP modulates plasma membrane phospholipid dynamics and facilitates cell surface retention of nutrient transporters, thus supporting nutrient uptake and impacting the inflammatory response (Yu et al., 2022 [↗](#)). Nutrient transport is key to T cell fate decisions and has been considered signal 4 to T cell fate choices (Chapman and Chi, 2022 [↗](#); Long et al., 2021 [↗](#)). The changes in ion channel related gene expression and MP level in *Nrn1*^{-/-} cells were accompanied by differential expression of AA transporter genes and nutrient sensing activity that impacted mTORC1 pathway activation and cellular glycolytic state (**Figures 3** [↗](#) and

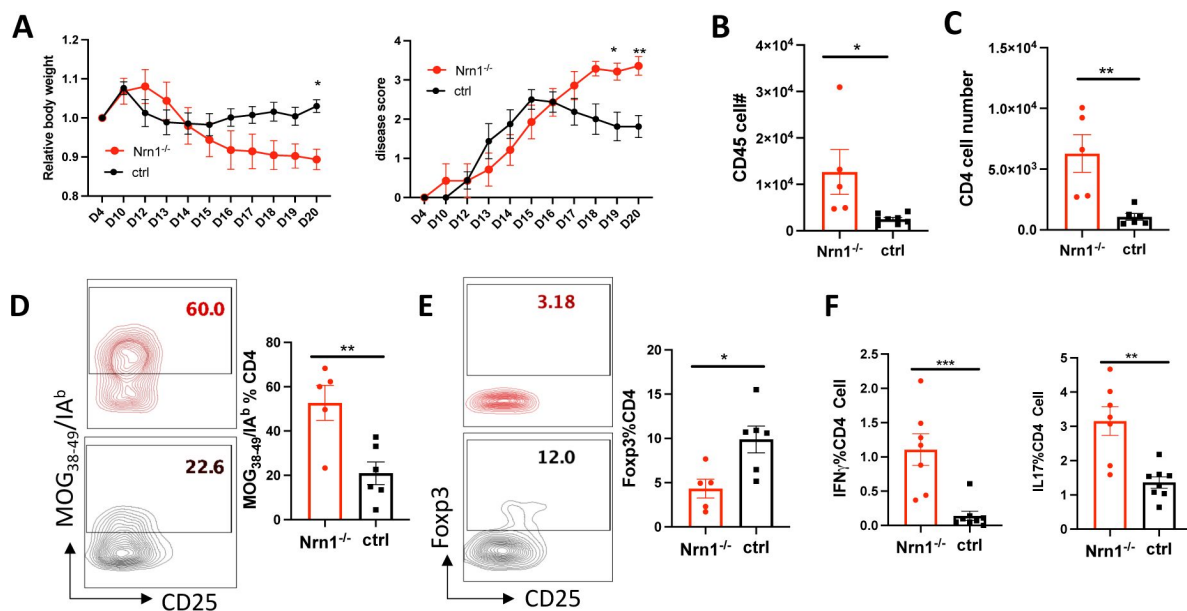


Figure 5.

Nrn1 deficiency exacerbates autoimmune EAE disease.

(A) Aggravated body weight loss and protracted EAE disease in $Nrn1^{-/-}$ mice. (B) CD45⁺ cell number in the spinal cord infiltrates. (C) CD4⁺ cell number in the spinal cord infiltrates. (D) MOG₃₈₋₄₉/IA^b tetramer staining of spinal cord infiltrating CD4 cells. (E) Foxp3⁺ proportion among CD4⁺ cells in spinal cord infiltrates. (F) IFN γ ⁺ and IL17⁺ cell proportion among CD4⁺ cells in draining lymph nodes. $n \geq 5$ mice per group. Data represent three independent experiments. The P value was calculated by 2way ANOVA for (A). The p-value was calculated by the unpaired student t-test for (B-F). * $P < 0.05$, ** $P < 0.01$.

4 [↗](#)). These results corroborate previous observations on the connection of MP in nutrient acquisition and metabolic change and support the role of *Nrn1* in coordinating T cell electric and metabolic adaptation (Yu et al., 2022 [↗](#)).

Although *Nrn1*, as a small GPI-anchored protein, does not have channel activity by itself, it has been identified as one of the components in the AMPAR complex (Pandya et al., 2018 [↗](#); Schwenk et al., 2012 [↗](#); Subramanian et al., 2019 [↗](#)). Na^+ -influx through the AMPA type ionotropic glutamate receptor can quickly depolarize the postsynaptic compartment and potentiate synaptic transmission in neurons. We have observed increased expression of AMPAR subunits in *Nrn1*^{-/-} iTreg and Te cells (Figure 3D [↗](#), Figure 4-figure supplement 1), implicating potential change in AMPAR activity in *Nrn1*^{-/-} under Treg and Te cell context. Glutamate secreted by proliferating cells may influence T cell function through AMPAR. High glutamate levels are detected at the autoimmune disease site and tumor interstitial fluid (Bonnet et al., 2020 [↗](#); McNearney et al., 2004 [↗](#); Sullivan et al., 2019 [↗](#)). Moreover, AMPAR has been implicated in exacerbating autoimmune disease (Bonnet et al., 2015 [↗](#); Sarchielli et al., 2007 [↗](#)). The increased expression of AMPAR subunits in *Nrn1*^{-/-} cells supports the potential connection of *Nrn1* and AMPAR and warrants future investigation on the possibility that *Nrn1* functions through AMPAR, impacting T cell electric change. Besides AMPAR, *Nrn1* has been reported to function through the insulin receptor and fibroblast growth factor pathway (Shimada et al., 2016 [↗](#); Yao et al., 2012 [↗](#)). Subramanian et al have suggested that rather than a traditional ligand with its cognate receptor, *Nrn1* may function as an adaptor to receptors to perform diverse cell-type-specific functions (Subramanian et al., 2019 [↗](#)). Our results do not rule out these possibilities.

Overall, we found that *Nrn1* expression in Treg and Te cells can impact cellular electric state, nutrient sensing, and metabolism in a cell context dependent manner. The predominant enrichment of ion channel related gene sets in both Treg and Te cell context underscores the importance of *Nrn1* in modulating ion balance and MP. The changes in ion channels and nutrient transporter expression in Treg and Te cells and associated functional consequences highlight the importance of *Nrn1* in coordinating cell metabolic changes through channels and transporters during the adaptive response and contribute to the balance of tolerance and immunity.

Materials and Methods

Mouse models

The *Nrn1*^{-/-} mice (Fujino et al., 2011 [↗](#)), Foxp3DTRGFP (FDG)(Kim et al., 2007 [↗](#)), and TCR α ^{-/-} mice were obtained from the Jackson Laboratory. OTII mice on Thy1.1⁺ background was kindly provided by Dr. Jonathan Powell. Rag2^{-/-} mice were maintained in our mouse facility. 6.5 TCR transgenic mice specific for HA antigen and C3HA mice (both on the B10.D2 background) have been described previously (Huang et al., 2004 [↗](#)). *Nrn1*^{-/-} mice were crossed with OTII mice to generate *Nrn1*^{-/-}_OTII⁺ mice, ctrl_OTII⁺ mice. *Nrn1*^{-/-} mice were also crossed with FDG mice to generate *Nrn1*^{-/-}_FDG and ctrl_FDG mice. All mice colonies were maintained in accordance with the guidelines of Johns Hopkins University and the institutional animal care and use committee

Antibodies and Reagents

We have used the following antibodies: Anti-CD3 (17A2), anti-CD4 (RM4-5), anti-CD8a (53-6.7), anti-CD25 (PC61), anti-CD45.1 (A20), anti-CD45.2 (104), anti-CD62L (MEL-14), CD73 (TY/111.8), anti-CD90.1 (OX-7), anti-CD90.2 (30-H12), anti-TCR V775.1, 5.2 (MR9-4), anti-PD1 (29F.1A12), anti-IFN γ (XMG1.2), anti-IL17 α (TC11-18H10.1), anti-TNF α (MP6-XT22), anti-Tbet (4B10), anti-Ki67 (16A8) were purchased from Biolegend. Anti-CD44 (IM7), CD45 (30-F11), anti-CD69(H1.2F3) were purchased from BD Bioscience. Anti-FOXP3 (FJK-16s) was purchased from eBioscience. The flow cytometry data were collected using BD Celesta (BD Biosciences) or Attune Flow Cytometers (ThermoFisher). Data were analyzed using FlowJo (Tree Star) software.

Mouse monoclonal anti-Nrn1 antibody (Ab) against Nrn1 was custom-made (A&G Pharmaceutical). The specificity of anti-Nrn1 Ab was confirmed by ELISA, cell surface staining of Nrn1 transfected 293T cells, and western blot of Nrn1 recombinant protein and brain protein lysate from WT mice or Nrn1^{-/-} mice (data not shown). OVA₃₂₃₋₃₃₉ peptide and MOG₃₅₋₅₅ was purchased from GeneScript. Incomplete Freund's adjuvant (IFA) and Mycobacterium tuberculosis H37Ra (killed and desiccated) were purchased from Difco. Pertussis toxin was purchased from List Biological Laboratories and diphtheria toxin was obtained from Millipore-Sigma.

Cell purification and culture

Naïve CD4 cells were isolated from the spleen and peripheral lymph node by a magnetic bead-based purification according to the manufacturer's instruction (Miltenyi Biotech). Purified CD4 cells were stimulated with plate-bound anti-CD3 (5ug/ml, Bio-X-Cell) and anti-CD28 (2ug/ml, Bio-X-Cell) for 3 days, in RPMI1640 medium supplemented with 10%FBS, HEPES, penicillin/streptomycin, MEM Non-Essential Amino Acids, and β-mercaptoethanol. For iTreg cell differentiation, cells were stimulated in the presence of human IL2 (100u/ml, PeproTech), human TGF² (10ng/ml, PeproTech), anti-IL4, and anti-IFNγ antibody (5ug/ml, Clone 11B11 and clone XMG1.2, Bio-X-Cell) in 10% RPMI medium. CD4⁺ Te cells were differentiated without additional cytokine or antibody for three days, followed by additional culture for 2 days in IL2 100u/ml in 10%RPMI medium. nTreg cells were isolated by sorting from the FDG CD4⁺ fraction based on Foxp3⁺GFP and CD25 expression (CD4⁺CD25⁺GFP⁺). Alternatively, nTreg cells were enriched from CD4 cells by positive selection using the CD4⁺CD25⁺ Regulatory T Cell Isolation Kit from Miltenyi.

Self-antigen induced tolerance model

1x10⁶ HA-specific Thy1.1⁺ 6.5 CD4 cells from donor mice on a B10.D2 background were transferred into C3-HA recipient mice, where HA is expressed as self-antigen in the lung; or into WT B10.D2 mice followed by infection with Vac-HA virus (1x10⁶ pfu). HA-reactive T cells were recovered from the lung-draining lymph node of C3-HA host mice or WT B10.D2 Vac-HA infected mice at indicated time points by cell sorting. RNA from sorted cells was used for qRT-PCR assay examining Nrn1 expression.

Peptide-induced T cell anergy model

5x10⁵ Polyclonal Treg cells from CD45.1⁺ C57BL/6 mice were mixed with 5x10⁶ thy1.1⁺ OTII cells from Nrn1^{-/-}_OTII or ctrl_OTII mice and transferred by *i.v.* injection into TCRα^{-/-} mice. 100ug of OVA₃₂₃₋₃₃₉ dissolved in PBS was administered *i.v.* on days 1, 4, and 7 after cell transfer. Host mice were harvested on day 13 after cell transfer, and cells from the lymph node and spleen were further analyzed.

In vivo Treg suppression assay

nTreg cells from CD45.2⁺CD45.1⁻ Nrn1^{-/-} or ctrl mice (5x10⁵/mouse) in conjunction with CD45.1⁺ splenocytes (2x10⁶/mouse) from FDG mice were cotransferred *i.p.* into Rag2^{-/-} mice. The CD45.1⁺ splenocytes were obtained from FDG mice pretreated with DT for 2 days to deplete Treg cells. Treg suppression toward CD45.1⁺ responder cells was assessed on day 7 post cell transfer. Alternatively, 7 days after cell transfer, Rag2^{-/-} hosts were challenged with an *i.d.* inoculation of B16F10 cells (1x10⁵). Tumor growth was monitored daily. Treg-mediated suppression toward anti-tumor response was assessed by harvesting mice day 18-21 post-tumor inoculation.

Induction of autoimmunity by transient Treg depletion

To induce autoimmunity in Nrn1^{-/-}_FDG and ctrl_FDG mice, 1ug/mouse of DT was administered *i.p.* for two consecutive days, and the weight loss of treated mice was observed over time.

EAE induction

EAE was induced in mice by subcutaneous injection of 200 µg MOG_{35–55} peptide with 500 µg M. tuberculosis strain H37Ra (Difco) emulsified in incomplete Freund Adjuvant oil in 200 µl volume into the flanks at two different sites. In addition, the mice received 400 ng pertussis toxin (PTX; List Biological Laboratories) *i.p.* at the time of immunization and 48 h later. Clinical signs of EAE were assessed daily according to the standard 5-point scale (Miller et al., 2007 [DOI](#)): normal mouse; 0, no overt signs of disease; 1, limp tail; 2, limp tail plus hindlimb weakness; 3, total hindlimb paralysis; 4, hindlimb paralysis plus 75% of body paralysis (forelimb paralysis/weakness); 5, moribund.

ELISA

MaxiSorp ELISA plates (ThermoScientific Nunc) were coated with 100 µl of 1 µg/ml anti-mIL-2 (BD Pharmingen #554424) at 4°C overnight. Coated plates were blocked with 200 µl of blocking solution (10% FBS in PBS) for 1 hr at room temperature (RT) followed by incubation of culture supernatant and mIL-2 at different concentrations as standard. After 1 hr, plates were washed and incubated with anti-mIL-2-biotin (BD Pharmingen #554426) at RT for 1 hr. After 1 hr, plates were incubated with 100 µl of horseradish peroxidase-labeled avidin (Vector Laboratory, #A-2004) 1 µg/ml for 30 min. After washing, samples were developed using the KPL TMB Peroxidase substrate system (Seracare #5120-0047) and read at 405 or 450 nm after the addition of the stop solution.

Quantitative RT-PCR

RNA was isolated using the RNeasy Micro Kit (Qiagen 70004) following the manufacturer's instructions. RNA was converted to cDNA using the High-Capacity cDNA Reverse Transcription Kit (ThermoFisher Scientific #4368814) according to the manufacturer's instructions. The primers of murine genes were purchased from Integrated DNA Technology (IDT). qPCR was performed using the PowerUp SYBR Green Master Mix (ThermoFisher Scientific #A25780) and the Applied Biosystems StepOnePlus 96-well real-time PCR system. Gene expression levels were calculated based on the Delta-Delta Ct relative quantification method. Primers used for Nrn1 PCR were as follows: GCGGTGCAAATAGCTTACCTG (forward); CGGTCTTGATGTTCTGCTTGTGTC (reverse).

Ca⁺⁺ flux and Membrane potential measurement

To measure Ca⁺⁺ flux, CD4 cells were loaded with Fluo4 dye at 2 µM in the complete cell culture medium at 37°C for 30 min. Cells were washed and resuspended in HBSS Ca⁺⁺ free medium and plated into 384 well glass bottom assay plate (minimum of 4 wells per sample). Ca⁺⁺ flux was measured using the FDSS6000 system (Hamamatsu Photonics). To measure store-operated calcium entry (SOCE), after the recording of the baseline T cells Ca⁺⁺ fluorescent for 1 min, thapsigargin (TG) was added to induce store Ca⁺⁺ depletion, followed by the addition of Ca⁺⁺ 2 µM in the extracellular medium to observe Ca⁺⁺ cellular entry.

Membrane potential was measured using FLIPR Membrane Potential Assay kit (Molecular devices) according to the manufacturer's instructions. Specifically, T cells were loaded with FLIPR dye by adding an equal volume of FLIPR dye to the cells and incubated at 37°C for 30 minutes. Relative membrane potential was measured by detecting FLIPR dye incorporation using flow cytometry.

Extracellular flux analysis (Seahorse assays)

Real-time measurements of extracellular acidification rate (ECAR) and oxygen consumption rate (OCR) were performed using an XFe-96 Bioanalyser (Agilent). T cells (2 × 10⁵ cells per well; minimum of four wells per sample) were spun into previously poly-d-lysine-coated 96-well plates (Seahorse) in complete RPMI-1640 medium. ECAR was measured in RPMI medium in basal condition and in response to 25 mM glucose, 1 µM oligomycin, and 50 mM of 2-DG (all from Sigma

Aldrich). OCR was measured in RPMI medium supplemented with 25mM glucose, 2mM L-glutamine, and 1mM sodium pyruvate, under basal condition and in response to 1μM oligomycin, 1.5μM of carbonylcyanide-4-(trifluoromethoxy)-phenylhydrazone (FCCP) and 1μM of rotenone and antimycin (all from Sigma Aldrich).

RNAseq and data analysis

RNAseq samples: 1. Anergic T cell analysis. Ctrl and *Nrn1*^{-/-} OTII cells were sorted from the host mice (n=3 per group). 2. iTreg cell analysis. In vitro differentiated *Nrn1*^{-/-} and ctrl iTreg cells were replated in resting condition (IL2 100u/ml) or stimulation condition (IL2 100u/ml and aCD3 5ug/ml). Cells were harvested 20 hr after replating for RNAseq analysis. 3. Effector T cells. *Nrn1*^{-/-} and ctrl CD4 Tn cells were activated for 3 days (aCD3 5ug/ml, aCD28 2ug/ml), followed by replating in IL2 medium (100u/ml). The cells were harvested two days after replating and subjected to RNAseq analysis.

RNA-sequencing analysis was performed by Admera Health (South Plainfield, NJ). Read quality was assessed with FastQC and aligned to the Mus Musculus genome (Ensembl GRCm38) using STAR aligner (version 2.6.0)([Dobin et al., 2013](#)). Aligned reads were counted using HTSeq (version 0.9.0)([Anders et al., 2015](#)), and the counts were loaded into R (The R Foundation). DESeq2 package (version 1.24.0)([Love et al., 2014](#)) was used to normalize the raw counts. GSEA was performed using public gene sets (HALLMARK, and GO)([Subramanian et al., 2005](#)). Cytoscape was used to display enriched gene sets cluster ([Shannon et al., 2003](#)).

Statistical analysis. All numerical data were processed using Graph Pad Prism 10. Data are expressed as the mean +/- the SEM, or as stated. Statistical comparisons were made using an unpaired student t-test or ANOVA with multiple comparison tests where 0.05 was considered significant, and a normal distribution was assumed. The p values are represented as follows: * p<0.05; ** p<0.01; *** p<0.001, **** p<0.0001.

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Author contributions

H. Y. was involved in all aspects of this study, including planning and performing experiments, analyzing and interpreting data, and writing the manuscript. H. N. and J. B. were involved in performing experiments and data interpretation. P. V., Y. Z., and A. L. analyzed *Nrn1* expression in Treg cells and carried out Treg suppression and functional assay. M. M. was involved in *Nrn1* expression, Treg suppression assay, and manuscript writing. C. H. and C.D. conducted the anergy

and Te cell differential gene expression study; Y.Z., J. F., and K. C. conducted an autoimmune inflammation study, and M. H. helped with mouse colony genotyping. X. Z. and Z. L. contributed to bioinformatic analysis. D.M. P. oversaw the project and was involved in data interpretation and manuscript preparation.

Competing interests

C.D. is a co-inventor on patents licensed from JHU to BMS and Janssen and is currently an employee of Janssen Research. D.M.P. is a consultant for Compugen, Shattuck Labs, WindMIL, Tempest, Immunai, Bristol-Myers Squibb, Amgen, Janssen, Astellas, Rockspring Capital, Immunomic, and Dracen; owns founders' equity in ManaT Bio Inc., WindMIL, Trex, Jounce, Enara, Tizona, Tieza, and RAPT; and receives research funding from Compugen, Bristol-Myers Squibb, and Enara. All other authors do not have conflicting financial interests.

Data availability

The raw RNA sequencing data has been deposited under the GEO accession no. GSE121908 and GSE224083.

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Reviewer #1 (Public Review):

The manuscript by Yu et al seeks to investigate the role of neuritin (Nr1h3), identified as a marker of anergic cells, in the biology of regulatory (Tregs) and conventional (Tconv) T cells. Although the role of Nr1h3 expressed by Tregs has already been explored (Gonzalez-Figueroa 2021 cited in the manuscript), this manuscript shows original new data suggesting that this molecule would be important in promoting Treg function and inhibiting Tconv effector function by acting at the level of membrane potential and molecule transport across the plasma membrane. However, it is disappointing that reading this manuscript leaves an impression of incomplete work done too quickly. Multiple models have been used, but none has been studied thoroughly enough to provide really conclusive and unambiguous data. For example, 5 different models were used to study T cells in vivo. It would have been preferable to use fewer, but to go further in the study of mechanisms. In the absence of a more in-depth study, the conclusions drawn by the authors are often open to question. Major points concern the fact that there are enough biological replicates for most experiments, some critical controls and data are lacking, and the authors have used iTregs rather than nTregs for many experiments (see below). This is unfortunate because the role of neuritin in T cell biology studied here is new and interesting.

Major points (in the order in which they appear in the text):

- (1) A real weakness of this work is the fact that in most of the results shown, there are few biological replicates with differences that are often small between Ctrl and Nr1h3^{-/-}. The systematic use of student's t-test may lead to thinking that the differences are significant, which is often misleading given the small number of samples, which makes it impossible to know whether the distributions are Gaussian and whether a parametric test can be used. RNAseq bulk data are based on biological duplicates, which is open to criticism.
- (2) The authors use Nr1h3^{+/+} and Nr1h3^{+/-} cells indiscriminately as control cells on the basis of similar biology between Nr1h3^{+/+} and Nr1h3^{+/-} cells at homeostasis. However, it is quite possible that the Nr1h3^{+/-} cells have a phenotype in situations of in vitro activation or in vivo inflammation (cancer, EAE). It would be important to discriminate Nr1h3^{+/-} and Nr1h3^{+/+} cells in the data or to show that both cell types have the same phenotype in these conditions too.
- (3) Figure 1A-D. Since the authors are using the Nr1h3 KO mice, it would be important to confirm the specificity of the anti-Nr1h3 mAb by FACS. Once verified, it would be important to add FACS results with this mAb in Figures 1A-C to have single-cell and quantitative data as well.
- (4) Figure 1E-H. The authors assume that this immunization protocol induces anergic cells, but they provide no experimental evidence for this. It would be useful to show that T cells are indeed anergic in this model, especially those that are OVA-specific. The lack of IL-2 production by Ctrl cells could be explained by the presence of fewer OVA-specific cells, rather than by an anergic status.

(5) Figure 2A-C and Figure 3. The use of iTregs to try to understand what is happening in vivo is problematic. iTregs are cells that have probably no equivalent in vivo, and so may have no physiological relevance. In any case, they are different from pTreg cells generated in vivo. Working with pTreg may be challenging, that is why I would suggest generating data with purified nTreg. Moreover, it was shown in the article of Gonzalez-Figueroa 2021 that *Nrn1*^{-/-} nTreg retained a normal suppressive function, which would not be what is concluded by the authors of this manuscript. Moreover, we do not even know what the % of Foxp3 cells is in the iTreg used (after differentiation and 20h of re-stimulation) and whether this % is the same between Ctlr and *Nrn1* KO cells.

(6) Figure 2D-L. The model is designed to study the role of *Nrn1* in nTreg. However, the % of Foxp3⁺ among CD45.2 nTreg cells fell to 5-15% of CD4⁺ cells (Figure 2F). Since we do not know what is the % of Foxp3 among the injected cells, we do not know whether this very low % is due to very high Treg instability or to preferential expansion of contaminating Tconvs. It is possible that the % of Tconv contaminant is high since Treg was sorted using beads and not FACS in some experiments. As it is very likely that there are Tconv contaminants that would be *Nrn1*^{-/-} in the group transferred with *Nrn1*^{-/-} "nTreg", the higher tumor rejection could be due to an overactivation of *Nrn1*^{-/-} Tconvs (rather than a defect in *Nrn1*^{-/-} Treg function).

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Reviewer #2 (Public Review):

Summary:

This manuscript explores the role of *Nrn1* in T cell tolerance. A previous study has demonstrated that *Nrn1* is up-regulated in the Tfr fraction of Foxp3⁺ T regulatory cells. These authors now confirm the expression of *Nrn1* in Tregs as well as report here that *Nrn1* is also greatly over-expressed in anergic CD4 T cells, and this is the stepping-off point for this investigation.

Most remarkably, experiments show that anergy induction is defective when T cells cannot express *Nrn1*. Furthermore, differentiation to a Foxp3⁺ Treg phenotype is inhibited in the absence of *Nrn1*, and the Tregs that do develop appear functionally defective. With such defects in the anergy induction and Treg differentiation and function, auto-reactive effector T cell activation is unrestrained, and *Nrn1*^{-/-} mice are more susceptible to severe EAE development.

Strengths:

The characterizations of T cell *Nrn1* expression both in vitro and in vivo are comprehensive and convincing. The in vivo functional studies of anergy development, Treg suppression, and EAE development are also well done to strengthen the notion that *Nrn1* is an important regulator of CD4 responsiveness.

Weaknesses:

The major weakness of this study stems from a lack of a clear molecular mechanism involving *Nrn1*. Previous studies of *Nrn1* have suggested its role as a soluble molecule involved in intracellular communication, perhaps influencing cellular ion channel function and/or triggering downstream NFAT and mTOR activation. However, a unique receptor for *Nrn1* has not been discovered and it remains unclear whether it acts in a cell-intrinsic or cell-extrinsic fashion for any particular cell type.

Data shown here provide evidence of alterations in the electrical and metabolic state of T cells when the *Nrn1* gene is deleted. *Nrn1*^{-/-} Tregs and Teffector cells each express a unique pattern of genes associated with Neurotransmitter receptor, Metal ion transmembrane transport, Amino acid transport, and mTORC1 signaling activities, different than that seen in wild-type mice. Although the biochemical and informatics studies are well-performed, it is my opinion that these results are inconclusive in part due to the absence of key "naive" control groups. This limits my ability to understand the significance of these data.

Specifically, studies of the electrical and metabolic state of *Nrn1*^{-/-} inducible Treg cells (iTregs) would benefit from similar data collected from wild-type and *Nrn1*^{-/-} naive CD4 T cells. Even though naive T cells don't express *Nrn1*, they may be positively influenced by soluble *Nrn1*. Does deletion of *Nrn1* lead to changes in metabolic and electrical state in naive T cells? Is that why *Nrn1* deletion in mice blocks naive T cell activation?

Since the loss of *Nrn1* inhibits the activation of T cells, are *Nrn1*^{-/-} iTregs transcriptionally, electrically, and metabolically similar to naive T cells due to their suboptimal activation? Does this account for their persistent functional defects? Or is up-regulation of *Nrn1* (and cell-intrinsic *Nrn1* signaling) necessary to complete Treg differentiation and to promote T regulatory function (similar to how cell-intrinsic *Nrn1* facilitates anergy induction)? The study of naive cells in parallel with iTregs would address these possibilities.

A comparison of *Nrn1*^{-/-} naive cells to Teffector cells should also be undertaken to reveal how it is that *Nrn1*^{-/-} Teffector cells regain the capacity to respond effectively to stimulation (e.g. increased mTOR activation) despite their early activation defects.

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Author response:

To Reviewer #1:

Thank you for your thorough review and comments on our work, which you described as "the role of neuritin in T cell biology studied here is new and interesting.". We have summarized your comments into two categories: biology and investigation approach, experimental rigor, and data presentation.

Biology and Investigation approach comments:

(1) Questions regarding the T cell anergy model:

Major point "(4) Figure 1E-H. The authors assume that this immunization protocol induces anergic cells, but they provide no experimental evidence for this. It would be useful to show that T cells are indeed anergic in this model, especially those that are OVA-specific. The lack of IL-2 production by Ctr cells could be explained by the presence of fewer OVA-specific cells, rather than by an anergic status."

T cell anergy is a well-established concept first described by Schwartz's group. It refers to the hyporesponsive T cell functional state in antigen-experienced CD4 T cells (Chappert and Schwartz, 2010; Fathman and Lineberry, 2007; Jenkins and Schwartz, 1987; Quill and Schwartz, 1987). Anergic T cells are characterized by their inability to expand and to produce IL2 upon subsequent antigen re-challenge. In this paper, we have borrowed the existing *in vivo* T cell anergy induction model used by Mueller's group for T cell anergy induction (Vanasek et al., 2006). Specifically, Thy1.1⁺ Ctrl or *Nrn1*^{-/-} TCR transgenic OTII cells were co-transferred with the congenically marked Thy1.2⁺ WT polyclonal Treg cells into TCR⁰^{-/-} mice. After anergy induction, the congenically marked TCR transgenic T cells were recovered by

sorting based on Thy1.1+ congenic marker, and subsequently re-stimulation *ex vivo* with OVA323-339 peptide. We evaluated the T cell anergic state based on OTII cell expansion *in vivo* and IL2 production upon OVA323-339 restimulation *ex vivo*.

“The authors assume that this immunization protocol induces anergic cells, but they provide no experimental evidence for this.”

Because the anergy model by Mueller's group is well established (Vanasek et al., 2006), we did not feel that additional effort was required to validate this model as the reviewer suggested. Moreover, the limited IL2 production among the control cells upon restimulation confirms the validity of this model.

“The lack of IL-2 production by Ctr cells could be explained by the presence of fewer OVA-specific cells, rather than by an anergic status”.

Cells from Ctrl and Nrn1^{-/-} mice on a homogeneous TCR transgenic (OTII) background were used in these experiments. The possibility that substantial variability of TCR expression or different expression levels of the transgenic TCR could have impacted IL2 production rather than anergy induction is unlikely.

Overall, we used this *in vivo* anergy model to evaluate the Nrn1^{-/-} T cell functional state in comparison to Ctrl cells under the anergy induction condition following the evaluation of Nrn1 expression, particularly in anergic T cells. Through studies using this anergy model, we observed a significant change in Treg induction among OTII cells. We decided to pursue the role of Nrn1 in Treg cell development and function rather than the biology of T cell anergy as evidenced by subsequent experiments.

Minor points “(6) On which markers are anergic cells sorted for RNAseq analysis?”

Cells were sorted out based on their congenic marker marking Ctrl or Nrn1^{-/-} OTII cells transferred into the host mice. We did not specifically isolate anergic cells for sequencing.

(2) Question regarding the validity of iTreg differentiation model.

Major point: “(5) Figure 2A-C and Figure 3. The use of iTregs to try to understand what is happening in vivo is problematic. iTregs are cells that have probably no equivalent in vivo, and so may have no physiological relevance. In any case, they are different from pTreg cells generated in vivo. Working with pTreg may be challenging, that is why I would suggest generating data with purified nTreg. Moreover, it was shown in the article of Gonzalez-Figueroa 2021 that Nrn1^{-/-} nTreg retained a normal suppressive function, which would not be what is concluded by the authors of this manuscript. Moreover, we do not even know what the % of Foxp3 cells is in the iTreg used (after differentiation and 20h of re-stimulation) and whether this % is the same between Ctrl and Nrn1 KO cells.”

We thank Reviewer #1 for their feedback. While it is true that iTregs made *in vitro* and *in vivo* generated pTregs display several distinctions (e. g., differences in Foxp3 expression stability, for example), we strongly disagree with this statement by Reviewer#1 “The use of iTregs to try to understand what is happening in vivo is problematic. iTregs are cells that have probably no equivalent in vivo, and so may have no physiological relevance.” The induced Treg cell (iTreg) model was established over 20 years ago (Chen et al., 2003; Zheng et al., 2002), and the model is widely adopted with over 2000 citations. Further, it has been instrumental in understanding different aspects of regulatory T cell biology (Hurrell et al., 2022; John et al., 2022; Schmitt and Williams, 2013; Sugiura et al., 2022).

Because we have observed reduced pTreg generation *in vivo*, we choose to use the *in vitro* iTreg model system to understand the mechanistic changes involved in Treg cell differentiation and function, specifically, neuritin's role in this process. We have made no claim that iTreg cell biology is identical to pTreg generated *in vivo* or nTreg cells. However, the iTreg culture system has proved to be a good *in vitro* system for deciphering molecular events involved in complex processes. As such, it remains a commonly used approach by many research groups in the Treg cell field (Hurrell et al., 2022; John et al., 2022; Sugiura et al., 2022). Moreover, applying the iTreg *in vitro* culture system has been instrumental in helping us identify the cell electrical state change in *Nrn1*^{-/-} CD4 cells and revealed the biological link between *Nrn1* and the ionotropic AMPA receptor (AMPA), which we will discuss in the subsequent discussion. It is technically challenging to use nTreg cells for T cell electrical state studies due to their heterogeneous nature from development in an *in vivo* environment and the effect of manipulation during the nTreg cell isolation process, which can both affect the T cell electrical state.

*“Moreover, it was shown in the article of Gonzalez-Figueroa 2021 that *Nrn1*^{-/-} nTreg retained a normal suppressive function, which would not be what is concluded by the authors of this manuscript.”*

We have also carried out nTreg studies *in vitro* in addition to iTreg cells. Similar to Gonzalez-Figueroa et al.'s findings, we did not observe differences in suppression function between *Nrn1*^{-/-} and WT nTreg using the *in vitro* suppression assay. However, *Nrn1*^{-/-} nTreg cells revealed reduced suppression function *in vivo* (Fig. 2D-L). In fact, Gonzalez-Figueroa et al. observed reduced plasma cell formation after OVA immunization in Treg-specific *Nrn1*^{-/-} mice, implicating reduced suppression from *Nrn1*^{-/-} follicular regulatory T (Tfr) cells. Thus, our observation of the reduced suppression function of *Nrn1*^{-/-} nTreg toward effector T cell expansion, as presented in Fig. 2D-L, does not contradict the results from Gonzalez-Figueroa et al. Rather, the conclusions of these two studies agree that *Nrn1* can play important roles in immune suppression observable *in vivo* that are not captured readily by the *in vitro* suppression assay.

*“Moreover, we do not even know what the % of Foxp3 cells is in the iTreg used (after differentiation and 20h of re-stimulation) and whether this % is the same between Ctrl and *Nrn1* KO cells.”*

We have stated in the manuscript on page 7 line 208 that “Similar proportions of Foxp3⁺ cells were observed in *Nrn1*^{-/-} and Ctrl cells under the iTreg culture condition, suggesting that *Nrn1* deficiency does not significantly impact Foxp3⁺ cell differentiation”. In the revised manuscript, we will include the data on the proportion of Foxp3⁺ cells before iTreg restimulation.

(3) Confirmation of transcriptomic data regarding amino acids or electrolytes transport change

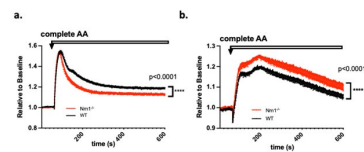
Minor point“(3) Would not it be possible to perform experiments showing the ability of cells to transport amino acids or electrolytes across the plasma membrane? This would be a more interesting demonstration than transcriptomic data.”

We appreciate Review# 1's suggestion regarding “perform experiments showing the ability of cells to transport amino acids or electrolytes across the plasma membrane”. We have indeed already performed such experiments corroborating the transcriptomics data on differential amino acid and nutrient transporter expression. Specifically, we loaded either iTreg or Th0 cells with membrane potential (MP) dye and measured MP level change after adding the complete set of amino acids (complete AA). Upon entry, the charge carried by AAs may

transiently affect cell membrane potential. Different AA transporter expression patterns may show different MP change patterns upon AA entry, as we showed in Author response image 1. We observed reduced MP change in *Nrn1*^{-/-} iTreg compared to the Ctrl, whereas in the context of Th0 cells, *Nrn1*^{-/-} showed enhanced MP change than the Ctrl. We can certainly include these data in the revised manuscript.

Author response image 1.

Membrane potential change induced by amino acids entry. **a.** *Nrn1*^{-/-} or WT iTreg cells loaded with MP dye and MP change was measured upon the addition of a complete set of AAs. **b.** *Nrn1*^{-/-} or WT Th0 cells loaded with MP dye and MP change was measured upon the addition of a complete set of AAs.



(4) EAE experiment data assessment

Minor point "(5) Figure 5F. How are cells re-stimulated? If polyclonal stimulation is used, the experiment is not interesting because the analysis is done with lymph node cells. This analysis should either be performed with cells from the CNS or with MOG restimulation with lymph node cells."

In the EAE study, the *Nrn1*^{-/-} mice exhibit similar disease onset but a protracted non-resolving disease phenotype compared to the WT control mice. Several reasons may contribute to this phenotype: 1. Enhanced T effector cell infiltration/persistence in the central nervous system (CNS); 2. Reduced Treg cell-mediated suppression to the T effector cells in the CNS; 3. Protracted non-resolving inflammation at the immunization site has the potential to continue sending T effector cells into CNS, contributing to persistent inflammation. Based on this reasoning, we examined the infiltrating T effector cell number and Treg cell proportion in the CNS. We also restimulated cells from draining lymph nodes close to the inflammation site, looking for evidence of persistent inflammation. When mice were harvested around day 16 after immunization, the inflammation at the local draining lymph node should be at the contraction stage. We stimulated cells with PMA and ionomycin intended to observe all potential T effector cells involved in the draining lymph node rather than only MOG antigen-specific cells. We disagree with Reviewer #1's assumption that "This analysis should either be performed with cells from the CNS or with MOG restimulation with lymph node cells.". We think the experimental approach we have taken has been appropriately tailored to the biological questions we intended to answer.

Experimental rigor and data presentation.

(1) Data labeling and additional supporting data

*Major points (2) The authors use *Nrn1*^{+/+} and *Nrn1*^{+/-} cells indiscriminately as control cells on the basis of similar biology between *Nrn1*^{+/+} and *Nrn1*^{+/-} cells at homeostasis. However, it is quite possible that the *Nrn1*^{+/-} cells have a phenotype in situations of *in vitro* activation or *in vivo* inflammation (cancer, EAE). It would be important to discriminate *Nrn1*^{+/-} and *Nrn1*^{+/+} cells in the data or to show that both cell types have the same phenotype in these conditions too.*

(3) Figure 1A-D. Since the authors are using the *Nrp1* KO mice, it would be important to confirm the specificity of the anti-Nrn1 mAb by FACS. Once verified, it would be important to add FACS results with this mAb in Figures 1A-C to have single-cell and quantitative data as well.

Minor points

(1) Line 119, 120 of the text. It is said that one of the most up-regulated genes in anergic cells is *Nrn1* but the data is not shown.

(2) For all figures showing %, the titles of the Y axes are written in an odd way. For example, it is written "Foxp3% CD4". It would be more conventional and clearer to write "% Foxp3+ / CD4+" or "% Foxp3+ among CD4+".

(4) For certain staining (Figure 3E, H) it would be important to show the raw data, in addition to MFI or % values.

We can adapt the labeling and provide additional data, including *Nrn1* staining on Treg cells and flow graphs for pmTOR and pS6 staining (Fig. 3H), as requested by Reviewer #1.

(2) Experimental rigor:

General comments:

"However, it is disappointing that reading this manuscript leaves an impression of incomplete work done too quickly."

We were discouraged to receive the comment, "this manuscript leaves an impression of incomplete work done too quickly." Our study of this novel molecule began without any existing biological tools such as antibodies, knockout mice, etc. Over the past several years, we have established our own antibodies for *Nrn1* detection, obtained and characterized *Nrn1* knockout mice, and utilized multiple approaches to identify the molecular mechanism of *Nrn1* function. Through the use of the *in vitro* iTreg system described in this manuscript, we identified the association of *Nrn1* deficiency with cell electrical state change, potentially connected to AMPAR function. We have further corroborated our findings by generating *Nrn1* and AMPAR T cell specific double knockout mice and confirmed that T cell specific AMPAR deletion could abrogate the phenotype caused by the *Nrn1* deficiency (see Author response image 2). We did not include the double knockout data in the current manuscript because AMPAR function has not yet been studied thoroughly in T cell biology, and we feel this topic warrants examination in its own right. However, the unpublished data support the finding that *Nrn1* modulates the T cell electrical state and, consequently, metabolism, ultimately influencing tolerance and immunity. In its current form, the manuscript represents the first characterization of the novel molecule *Nrn1* in anergic cells, Tregs, and effector T cells. While this work has led to several exciting additional questions, we disagree that the novel characterization we have presented is incomplete. We feel that our present data set, which squarely highlights *Nrn1*'s role as an important immune regulator while shedding unprecedented light on the molecular events involved, will be of considerable interest to a broad field of researchers.

"Multiple models have been used, but none has been studied thoroughly enough to provide really conclusive and unambiguous data. For example, 5 different models were used to study T cells in vivo. It would have been preferable to use fewer, but to go further in the study of mechanisms."

We have indeed used multiple *in vivo* models to reveal Nrn1's function in Treg differentiation, Treg suppression function, T effector cell differentiation and function, and the overall impact on autoimmune disease. Because the impact of ion channel function is often context-dependent, we examined the biological outcome of Nrn1 deficiency in several *in vivo* contexts. We would appreciate it if Reviewer#1 would provide a specific example, given the Nrn1 phenotype, of how to proceed deeper to investigate the electrical change in the *in vivo* models.

"Major points (1) A real weakness of this work is the fact that in most of the results shown, there are few biological replicates with differences that are often small between Ctrl and Nrn1 -/-. The systematic use of student's t-test may lead to thinking that the differences are significant, which is often misleading given the small number of samples, which makes it impossible to know whether the distributions are Gaussian and whether a parametric test can be used. RNAseq bulk data are based on biological duplicates, which is open to criticism."

We respectfully disagree with Reviewer #1 on the question of statistical power and significance to our work. We have used 5-8 mice/group for each *in vivo* model and 3-4 technical replicates for the *in vitro* studies, with a minimum of 2-3 replicate experiments. These group sizes and replication numbers are in line with those seen in high-impact publications. While some differences between Ctrl and Nrn1-/- appear small, they have significant biological consequences, as evidenced by the various Nrn1-/- *in vivo* phenotypes. Furthermore, we believe we have subjected our data to the appropriate statistical tests to ensure rigorous analysis and representation of our findings.

To Reviewer #2.

We thank Reviewer #2 for the careful review of the manuscript. We especially appreciate the comments that "The characterizations of T cell Nrn1 expression both *in vitro* and *in vivo* are comprehensive and convincing. The *in vivo* functional studies of anergy development, Treg suppression, and EAE development are also well done to strengthen the notion that Nrn1 is an important regulator of CD4 responsiveness."

"The major weakness of this study stems from a lack of a clear molecular mechanism involving Nrn1."

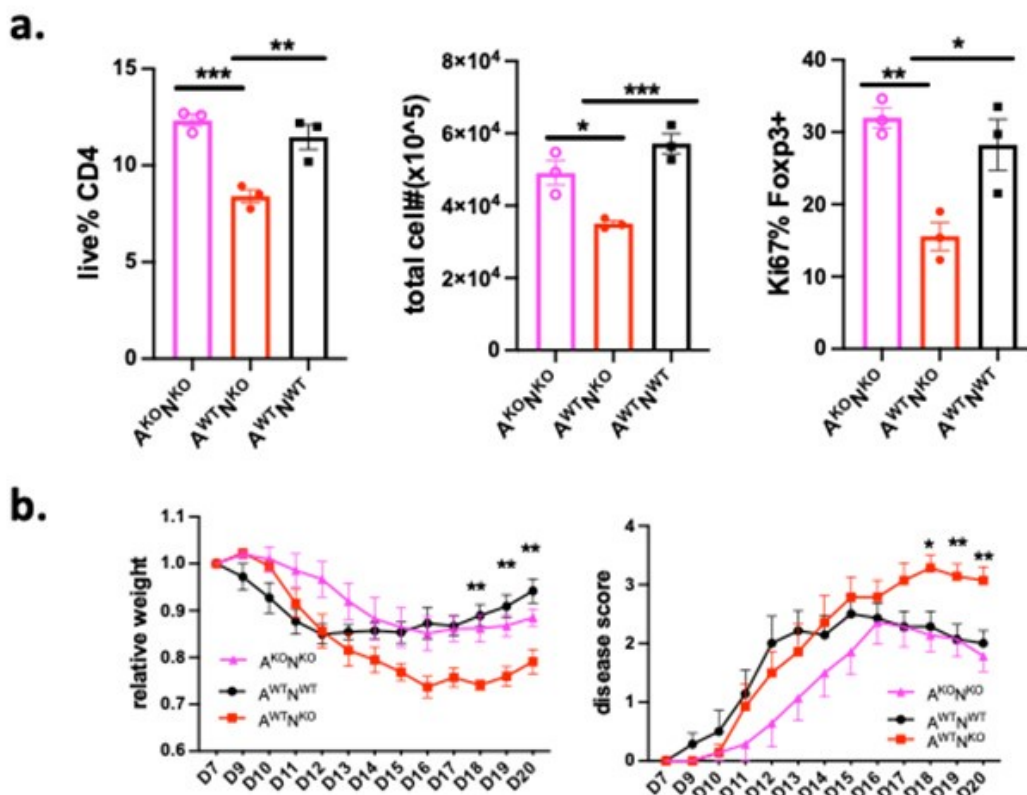
We fully understand this comment from Reviewer #2. The main mechanism we identified contributing to the functional defect of Nrn1-/- T cells involves novel effects on the electric and metabolic state of the cells. Although we referenced neuronal studies that indicate Nrn1 is the auxiliary protein for the ionotropic AMPA-type glutamate receptor (AMPA) and may affect AMPAR function, we did not provide any evidence in this manuscript as the topic requires further in-depth study.

For the benefit of this discussion, we include our preliminary Nrn1 and AMPAR double knockout data (Author response image 2), which indicates that abrogating AMPAR expression can compensate for the defect caused by Nrn1 deficiency *in vitro* and *in vivo*. This preliminary data supports the notion that Nrn1 modulates AMPAR function, which causes changes in T cell electric and metabolic state, influencing T cell differentiation and function.

Author response image 2.

Deletion of AMPAR expression in T cells compensates for the defect caused by Nrn1 deficiency. Nrn1-/- mice were crossed with T cell-specific AMPAR knockout mice (AMPA^{fl}/flCD4Cre⁺) mice. The following mice were generated and used in the experiment: T cell specific AMPAR-knockout and Nrn1 knockout mice (AKONKO), Nrn1 knockout mice

(AWTNKO), Ctrl mice (AWTNWT). **a.** Deletion of AMPAR compensates for the iTreg cell defect observed in *Nrn1*^{-/-} CD4 cells. iTreg live cell proportion, cell number, and Ki67 expression among Foxp3⁺ cells 3 days after aCD3 restimulation. **b.** Deletion of AMPAR in T cells abrogates the enhanced autoimmune response in *Nrn1*^{-/-} Mouse in the EAE disease model. Mouse relative weight change and disease score progression after EAE disease induction.



Ion channels can influence cell metabolism through multiple means (Vaeth and Feske, 2018; Wang et al., 2020). First, ion channels are involved in maintaining cell resting membrane potential. This electrical potential difference across the cell membrane is essential for various cellular processes, including metabolism (Abdul Kadir et al., 2018; Blackiston et al., 2009; Nagy et al., 2018; Yu et al., 2022). Second, ion channels facilitate the movement of ions across cell membranes. These ions are essential for various metabolic processes. For example, ions like calcium (Ca²⁺), potassium (K⁺), and sodium (Na⁺) play crucial roles in signaling pathways that regulate metabolism (Kahlfuss et al., 2020). Third, ion channel activity can influence cellular energy balance due to ATP consumption associated with ion transport to maintain ion balances (Erecińska and Dagani, 1990; Gerkau et al., 2019). This, in turn, can impact processes like ATP production, which is central to cellular metabolism. Thus, ion channel expression and function determine the cell's bioelectric state and contribute to cell metabolism (Levin, 2021).

Because the AMPAR function has not been thoroughly studied using a genetic approach in T cells, we do not intend to include the double knockout data in this manuscript before fully characterizing the T cell-specific AMPAR knockout mice.

"Although the biochemical and informatics studies are well-performed, it is my opinion that these results are inconclusive in part due to the absence of key "naive" control groups. This limits my ability to understand the significance of these data."

Specifically, studies of the electrical and metabolic state of *Nrn1*^{-/-} inducible Treg cells (iTregs) would benefit from similar data collected from wild-type and *Nrn1*^{-/-} naïve CD4 T cells."

We appreciate the reviewer's comments. This comment reflects two concerns in data interpretation:

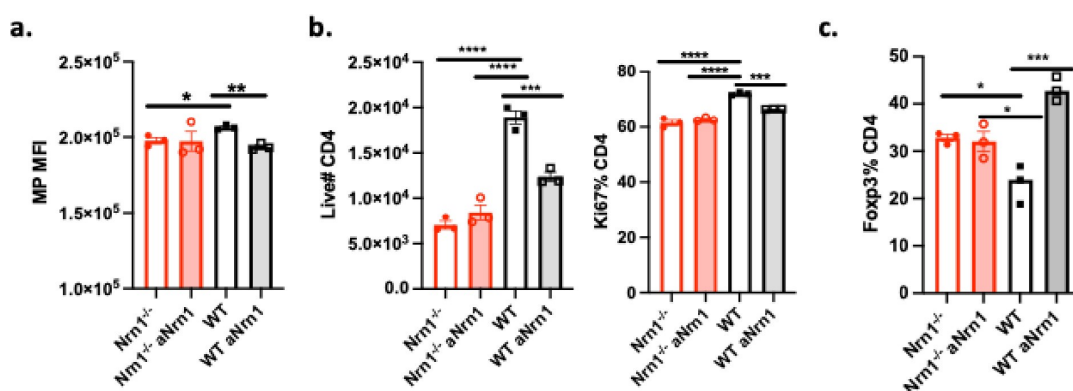
(1) Are *Nrn1*^{-/-} naïve T cells fundamentally different from WT cells? Does this fundamental difference contribute to the observed electrical and metabolic phenotype in iTreg or Th0 cells? This is a very good question we will perform the experiments as the reviewer suggested. While *Nrn1* is expressed at a basal (low) level in naïve T cells, deletion of *Nrn1* may cause changes in naïve T cell phenotype.

(2) Is the *Nrn1*^{-/-} phenotype caused by *Nrn1* functional deficiency or due to the secondary effect of *Nrn1* deletion, such as non-physiological cell membrane structure changes?

We have done the following experiment to address this concern. We have cultured WT T cells in the presence of *Nrn1* antibody and compared the outcome with *Nrn1*^{-/-} iTreg cells (Author response image 3). WT iTreg cells under antibody blockade exhibited similar changes as *Nrn1*^{-/-} iTreg cells, confirming the physiological relevance of the *Nrn1*^{-/-} phenotype.

Author response image 3.

Nrn1 antibody blockade in WT iTreg cell culture caused similar phenotypic change as in *Nrn1*^{-/-} iTreg cells. *Nrn1*^{-/-} and WT CD4 cells were differentiated under iTreg condition in the presence of anti-*Nrn1* (a*Nrn1*) antibody or isotype control for 3 days. Cells were restimulated with anti-CD3 and in the presence of a*Nrn1* or isotype. a. MP measured 18hr after anti-CD3 restimulation. b. live CD4 cell number and proportion of Ki67 expression among live cells three days after restimulation. c. The proportion of Foxp3⁺ cells among live cells three days after restimulation.



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